Master in Intelligent Interactive Systems
Universitat Pompeu Fabra

Adversarial Debiasing in Resume Screening for ICT Positions

Matteo Giorgi

Supervisor: Carlos Castillo

Co-Supervisor:

July 2025

Universitat
Pompeu Fabra
Barcelona

Master in Intelligent Interactive Systems
Universitat Pompeu Fabra

Adversarial Debiasing in Resume Screening for ICT Positions

Matteo Giorgi

Supervisor: Carlos Castillo
Co-Supervisor:

July 2025

Universitat Pompeu Fabra
Barcelona

Contents

Acknowledgement

I would like to express my sincere gratitude to my supervisor, Carlos, for their guidance, insights, and kindness throughout this work. A great thanks also goes to Anna for the great help she gave in completing this project. Thanks to their mentorship, I was able to explore the world of algorithmic fairness, a fascinating world that I truly hope to get to explore further.

I am also deeply grateful to my family, for their loving support and encouragement.

A special thought goes to my friends for being there throughout this journey, thank you Beli.

Their collective support made this work possible and meaningful. Thank you.

Abstract

AI hiring tools promise objectivity, but may deliver bias at scale. While algorithmic resume screening outperforms manual review by processing thousands of applications daily, these systems can perpetuate gender discrimination through learned patterns embedded in the data. This research challenges present data protection practices, suggesting that anonymization is insufficient against AI inference capabilities, and proposes an alternative approach: Adversarial Debiasing.

Our three-phase experimental pipeline investigated whether language models can detect gender from anonymized resumes, gender bias extent in ICT classification, and adversarial debiasing effectiveness. Using the FINDHR dataset of donated resumes with protected attributes and LiveCareer dataset for pre-training, we analyzed gender predictability from anonymized data, implemented transfer learning, then applied adversarial training using gradient reversal architecture to enforce gender uninformativeness in ICT features.

This research makes two primary contributions to fair AI in algorithmic hiring. First, we provide empirical evidence that standard anonymization fails against AI inference, with DistilBERT achieving 63.86% gender detection accuracy on anonymized resumes through systematic linguistic patterns in professional self-presentation. Second, we demonstrate adversarial debiasing effectiveness, achieving near-perfect gender uninformativeness, decreasing predictability from 63.10% to 51.19% (effectively random levels) while preserving 97.6% of ICT classification performance. Our lambda parameter optimization enabled transparent control over fairness-utility trade-offs.

These results have strong regulatory and industry implications. Current anonymization standards appear inadequate against AI pattern detection capabilities, demanding updated privacy protection frameworks. Our adversarial debiasing approach offers promise by embedding fairness constraints within feature learning processes, addressing proxy discrimination at its source while maintaining operational utility.

Keywords: Algorithmic bias; Resume screening; Adversarial debiasing; Fair machine learning; Gender discrimination

Chapter 1

Introduction

Modern recruitment processes are fundamentally shaped by the sheer volume of job applications, a challenge that has rendered traditional manual review methods inefficient and unsustainable [1, 2].

This operational strain has driven the widespread adoption of algorithmic resume screening systems, which promise to enhance efficiency, accelerate candidate processing, and offer unparalleled scalability [1, 2, 3, 4]. However, the integration of artificial intelligence (AI) into hiring, while addressing critical operational needs, has introduced an unforeseen and complex challenge: algorithmic bias [1, 5]. These AI systems, often trained on historical data reflecting past human prejudices, can inadvertently perpetuate or even amplify existing inequalities in hiring practices, presenting significant ethical, legal, and operational implications for organizations [6, 7].

To mitigate gender bias in algorithmic hiring systems, the present study explored Adversarial Debiasing, which leverages the same machine learning principles that can embed bias to actively combat it [8]. This approach employs adversarial training frameworks where models learn to make hiring-relevant predictions while simultaneously being trained to ignore protected attributes such as gender. By creating competing objectives within the learning process, our adversarial model forces the algorithm to develop representations that are informative for job classification but uninformative for demographic prediction, offering a principled approach to achieving fairness without sacrificing predictive performance.

This research addresses three fundamental questions that collectively examine the detection, manifestation, and mitigation of gender bias in automated ICT resume screening, focusing on this sector given its well-documented gender disparities and role as a developer of AI hiring tools. First, we investigate whether modern language models can detect protected attributes from anonymized resume text, challenging current privacy protection practices. Second, we quantify the extent of gender bias in ICT versus non-ICT classification systems. Finally, we evaluate adversarial debiasing effectiveness in reducing gender bias while maintaining classification performance.

1.1 Motivation

The primary motivations for this research encompass both the practical necessities of modern recruitment and the critical need to address emerging fairness concerns in automated systems.

1.1.1 The Modern Job Hiring Landscape

Major corporations today face an unprecedented influx of job applications that makes manual resume review increasingly impractical and inefficient[2]. Recent industry data indicates that each job posting attracted an average of 26 applicants in late 2022, marking a 23.8% increase from the previous quarter, with major corporations receiving thousands of applications for a single position[2, 9].

This volume challenge is compounded by present competitive market dynamics, where 70% of HR professionals describe the market as highly competitive[1]. This environment, coupled with high application volumes, creates immense pressure on recruitment processes to be both rapid and accurate. While only 20-30% of applications receive thorough review, 60% of talent acquisition professionals cite difficulty in finding candidates with precise skills and experience as their primary challenge[2].

This inefficiency translates to a 40% increase in recruitment bottlenecks and financial costs averaging $1,000-$1,500 per hire, with total hiring costs potentially reaching 3 to 4 times the position's salary[2].

The practical implications of this volume are severe: recruiters typically spend between 1 to 3 minutes per resume. When scaled to thousands of applications, this can extend the time to fill a position to between 4 and 6 weeks[2].

Thus, these operational challenges have made the adoption of algorithmic resume screening systems not only convenient but essential for organizational survival in the modern talent market. AI-powered systems have delivered transformative efficiency gains by processing thousands of resumes in minutes and reducing screening time by approximately 70% compared to traditional manual methods[1, 2]. Technology markets such as HR are experiencing substantial growth, driven primarily by the increasing embrace of cloud-based solutions and automation tools, with algorithmic resume screening standing as a foundational component of this digital transformation [10, 11].

However, the promise of objectivity that initially drove AI adoption has proven to be more complex than anticipated. While these systems excel at processing large volumes of data and maintaining consistency, they are demonstrably not immune to biases embedded in their training data and design choices [1, 12].

The Amazon hiring tool case study serves as a prominent example: the system, trained on 10 years of historical data predominantly reflecting male candidates in technical roles, learned to penalize resumes containing gender-associated terms such as "women's" in "captain of the women's soccer team" [13]. This case illustrates how AI systems, rather than eliminating human bias, can systematize and amplify existing discriminatory patterns at unprecedented scale.

1.1.2 Protected Attribute Inference and the Efficiency-Fairness Trade-off

Recent research demonstrates a critical vulnerability in current anonymization practices: modern machine learning systems can infer sensitive personal attributes from seemingly neutral CV data with concerning accuracy [14]. Even when explicit markers such as names, gender indicators, and age are removed, latent patterns in language use, employment gaps, educational backgrounds, and professional trajectories can serve as proxies that allow algorithms to deduce protected characteristics[14, 7]. This phenomenon undermines General Data Protection Regulation (GDPR) compliance foundations and creates legal risks, as organizations may unknowingly process protected attribute data through inference[15, 6]. However, simply abandoning AI in hiring is neither practical nor desirable given the operational realities of modern recruitment. Instead, the challenge lies in developing approaches that harness AI's efficiency and scalability while actively detecting and mitigating bias[6, 1]. The present research addresses this fundamental tension between the operational necessity of algorithmic hiring systems and the ethical imperative to ensure fair and equitable treatment of all candidates.

As AI systems become more sophisticated and widely adopted, the window for implementing effective bias detection and mitigation strategies narrows.

Our investigation focuses specifically on adversarial debiasing techniques as a promising approach for achieving this dual imperative, exploring how gradient reversal layers and adversarial training can be employed to reduce bias while maintaining classification performance[6, 16]. This technical approach is motivated by the recognition that bias mitigation must be built into AI systems at the architectural level, rather than applied as post-hoc corrections to discriminatory outputs[6].

1.2 Research Questions

Building upon the motivations outlined above, this research addresses three fundamental questions that collectively examine the detection, manifestation, and mitigation of bias in algorithmic resume screening systems.

Question 1: Protected Attribute Detectability and Anonymization Effectiveness.

Can machine learning models detect protected attributes from resume

text, and how does anonymization affect this capability?

This question investigates the fundamental vulnerability in current privacy protection practices. Given that anonymization is widely considered the primary defense against protected attribute discrimination in hiring, understanding its effectiveness against modern AI systems is crucial. The question examines whether state-of-the-art language models can infer sensitive characteristics such as gender from resume content even after explicit identifiers have been removed.

This investigation directly addresses the concept of privacy implications [14] and provides empirical evidence for evaluating current GDPR compliance strategies in algorithmic hiring systems.

Question 2: Bias Manifestation in ICT Classification.
What is the extent of gender bias in automated ICT vs non-ICT classification systems?

This question focuses on quantifying bias in an application domain where historical gender disparities are well-documented: The Information and Communication Technology (ICT) sector.

Rather than examining bias in abstract terms, this research question specifically measures how automated classification systems reproduce or amplify existing gender imbalances in ICT hiring. The investigation examines whether the probability of being classified as suitable for ICT positions varies systematically based on candidate gender, even when technical qualifications and experience levels are comparable.

Question 3: Adversarial Debiasing Effectiveness.
How effective is adversarial debiasing in reducing gender bias while maintaining classification performance?

This question evaluates a specific technical approach to bias mitigation, examining the practical viability of adversarial training methods in real-world hiring scenarios. The research investigates whether gradient reversal techniques can successfully force classification models to make decisions independent of gender information while preserving their ability to accurately distinguish between ICT and non-ICT candidates.

The question addresses the fundamental trade-off between fairness and accuracy that characterizes most bias mitigation approaches, providing empirical evidence for whether adversarial methods can achieve meaningful bias reduction without compromising the operational utility that drives AI adoption in recruitment.

1.3 Contributions and Scope

This research makes two primary contributions to the field of fair AI in algorithmic hiring:

Empirical Analysis of Protected Attribute Detectability:
We provide a systematic evaluation of modern language models' ability to infer protected attributes from anonymized resume text, achieving F1 scores of 60-65% for gender detection. This finding demonstrates critical vulnerabilities in current anonymization practices and establishes the inadequacy of existing privacy protection measures in GDPR-compliant hiring systems.

Adversarial Debiasing for Fair ICT Classification with respect to gender:
We implement and evaluate gradient reversal-based adversarial training for bias mitigation in resume screening, providing empirical evidence of its effectiveness in reducing gender bias while maintaining classification utility. This represents one of the first applications of adversarial debiasing techniques to real resume data with verified protected attribute labels.

1.3.1 Scope:

This research focuses specifically on gender bias in ICT vs non-ICT classification for several strategic reasons:

- Domain Significance: The ICT sector exhibits well-documented gender disparities (17% female representation [17]) and serves as a primary developer of

AI hiring tools, making it a critical domain for bias investigation with broad implications for AI fairness across industries.

- Technical Feasibility: Binary gender classification and ICT categorization provide a controlled experimental environment for isolating bias effects while maintaining sufficient statistical power for meaningful analysis with available datasets.

- Methodological Clarity: This focused scope enables the evaluation of adversarial debiasing techniques without the confounding variables introduced by multi-class protected attributes or complex job taxonomies.

Notably, this research does not address non-binary gender identities or intersectional bias effects, nor does it investigate protected attributes beyond gender. Furthermore, our scope does not include fine-grained job classification beyond the ICT/non-ICT binary or bias mitigation techniques other than adversarial training. These design choices were made for reasons of time and methodological simplicity, and we discuss them further at the end of this research. We believe integrating these aspects into future work represents an important direction.

1.4 Data Protection and Ethical Considerations

When developing algorithmic CV analysis systems, particular attention must be paid to data protection regulations, specifically the General Data Protection Regulation (GDPR) and the recently enacted AI Act (August 2024). The intersection of these frameworks presents both opportunities and challenges for bias detection research [18].

This research leverages a unique dataset of anonymized CVs paired with voluntary survey responses regarding protected characteristics, including both special and non-special GDPR category data.

1.4.1 Regulatory Framework and Compliance

The AI Act's Article 10(5) specifically permits processing of special categories of personal data "to the extent strictly necessary for the purposes of ensuring bias monitoring, detection and correction" in high-risk AI systems, provided appropriate safeguards are implemented [18]. However, this allowance must be reconciled with GDPR's more restrictive provisions under Article 9, which generally prohibits processing of sensitive data categories including racial or ethnic origin, political opinions, and data concerning sex life or sexual orientation [15].

The present research operates within this regulatory framework by utilizing only voluntarily provided survey data with explicit participant consent and implementing data minimization principles by processing only attributes necessary for bias detection.

1.4.2 Ethical Approval and Data Handling Protocols

All data processing strictly followed ethical research protocols. While the dataset contains directly identifying information, all CVs were pre-anonymized by removing names, contact details, and explicit demographic markers. Survey responses, which link protected attributes to resume content, were collected under informed consent, with participants fully aware their data would be used for algorithmic bias research. We operated under a Non-Disclosure Agreement, ensuring the dataset remains confidential and will not be disclosed, though no encryption was used during the procedure.

Chapter 2

Related Work and Theoretical Background

2.1 Algorithmic Bias in Hiring

The automatic analysis of resumes represents a growing segment of the broader algorithmic hiring ecosystem, where employers are increasingly adopting this technology throughout the recruitment pipeline [19]. Such technological proliferation has made algorithmic fairness especially applicable in the hiring domain due to its high stakes and structural inequalities, yet the discourse around these technologies often oscillates between two extremes: the optimistic view that algorithms can replace biased human decision-making, and the pessimistic perspective that they merely automate discrimination [19, 20].

2.1.1 Current Technological Landscape and Bias Mechanisms

Hiring is rarely a single decision point but rather "a cumulative series of small decisions" where predictive technologies play varied roles throughout the hiring funnel [21]. While algorithmic tools rarely make affirmative hiring decisions, they frequently automate rejections, particularly in the early stages of the hiring process. This pattern is especially relevant to resume screening, where initial filtering can systematically exclude candidates based on criteria that may correlate with protected characteristics [21].

Modern employers use predictive tools at every stage: from determining who sees job advertisements through digital sourcing platforms, to estimating applicant performance, to forecasting candidate salary requirements [21, 22].

The manifestation of bias in algorithmic hiring systems stems from multiple sources that are particularly pronounced in resume screening applications. Recent studies show insights that simply removing sensitive characteristics from data is insufficient to eliminate bias [19, 20, 23]. Resume screening algorithms trained on historical hiring decisions can reproduce systemic patterns of inequity even when explicitly ignoring protected attributes such as gender, age, or ethnicity. This occurs because seemingly neutral features like educational institution, previous employers, or even language patterns can serve as proxies for protected attributes [24].

Recent empirical evidence demonstrates the persistence of these proxy effects in resume screening contexts. Studies show that applicants with "white-sounding names" receive 50% more callbacks than those with "African-American sounding names," even when qualifications are identical, indicating that bias operates through multiple textual cues beyond explicit demographic markers [22].

2.1.2 Resume Screening Versus Other Hiring Tools

Resume screening occupies a unique position within the algorithmic hiring landscape due to its reliance on unstructured text data and its function as an initial filtering mechanism. Unlike structured assessment tools that evaluate specific skills through standardized tests, resume screening systems must extract meaningful signals from highly variable textual content [22, 9, 25]. This creates distinct bias vulnerabilities:

Keyword Dependency: Traditional resume screening tools rely heavily on keyword matching, which can systematically disadvantage candidates who use different terminology or lack access to insider knowledge about effective resume formatting [26, 22]. This approach may particularly impact candidates from non-traditional educational backgrounds or those transitioning between career fields.

Format Bias: Automated systems often struggle with non-standard resume formats, potentially creating technological barriers that correlate with socioeconomic status or cultural background [26, 22]. Candidates with access to professional resume formatting services or specific software may gain systematic advantages unrelated to job qualifications.

Contextual Limitations: Unlike human reviewers who can interpret career narratives and non-linear professional paths, automated resume screening often fails to recognize the value of diverse experiences, potentially penalizing candidates with non-traditional career trajectories [22].

2.1.3 Protected Attribute Inference from Text

A critical development in understanding bias mechanisms is the growing evidence of protected attribute inference capabilities in modern language models. Research demonstrates that sophisticated AI systems can deduce sensitive characteristics from text with alarming precision, reaching up to 85.5% accuracy on traits like age or gender, with accuracy remaining problematically high even after anonymization processing [27]. The inference mechanisms operate through subtle linguistic patterns, educational indicators, employment gap patterns, and other textual cues that correlate with protected attributes [24]. For instance, employment gaps may disproportionately affect women due to caregiving responsibilities, while certain linguistic styles or institutional affiliations may correlate with ethnic or socioeconomic background. These patterns enable AI systems to effectively "see through" anonymization efforts, creating covert pathways for discriminatory outcomes.

2.1.4 Systemic Implications and Amplification Effects

Some hiring tools "assess, score, and rank jobseekers, sometimes overstating marginal or unimportant distinctions between similarly qualified candidates" [21]. This artificial precision in candidate evaluation can disproportionately impact members of underrepresented groups by amplifying small differences that may reflect systemic advantages rather than job-relevant qualifications. The numerical scores and rankings produced by resume screening systems may influence recruiters more than anticipated, creating a veneer of objectivity that masks underlying bias.

The challenge is further complicated by a lack of contextualized understanding in current approaches. The relationship between statistical measures of fairness and legal or ethical conceptions of non-discrimination remains contested, particularly in the context of civil rights laws applicable to employment. Whether, and more importantly, what types of algorithmic hiring can be less biased and more beneficial to society than low-tech alternatives currently remains unanswered, to the detriment of trustworthiness [19].

This uncertainty is particularly problematic for resume screening systems, which operate at scale and often with limited transparency. The "black box" nature of many AI-driven screening tools makes it difficult for candidates to understand rejection decisions and for employers to verify that their systems operate fairly across different demographic groups. This opacity conflicts with growing demands for algorithmic accountability and may expose organizations to legal and reputational risks as regulatory scrutiny of AI hiring tools intensifies [24].

2.2 Fairness in Machine Learning

Fairness in machine learning represents a complex socio-technical challenge that extends beyond algorithmic solutions to encompass fundamental questions of justice, equity, and societal values. This section establishes the theoretical foundations necessary for understanding bias detection and mitigation in resume screening systems.

2.2.1 Definitions of Fairness

Machine learning fairness can be broadly categorized into two primary paradigms: **Group Fairness** and **Individual Fairness**, each reflecting different philosophical approaches to achieving equitable outcomes.

Group Fairness seeks to ensure that different predefined groups are treated equitably, with outcomes distributed proportionally across demographic categories defined by protected attributes such as race, gender, age, and socioeconomic status. Key metrics include [28]:

- **Statistical Parity (Demographic Parity)**: Requires equal likelihood of positive classification across groups, formally expressed as $P(\hat{Y}=1|A=\text{minority}) = P(\hat{Y}=1|A=\text{majority})$. This metric ensures proportional representation in outcomes regardless of individual qualifications.
- **Equalized Odds**: A more stringent criterion demanding equal true positive rates (TPR) and false positive rates (FPR) across protected groups, ensuring that both correct and incorrect positive predictions occur at identical rates for all demographics.
- **Equality of Opportunity**: A relaxed version of equalized odds focusing solely on equal true positive rates, ensuring that qualified individuals from different groups have equal chances of positive outcomes.

Individual Fairness operates on the principle that similar individuals should receive similar treatment regardless of group membership. It's formal definition states that a model achieves individual fairness when inputs deemed "sufficiently close" according to an appropriate distance metric produce correspondingly close outputs [29]. However, this approach faces significant practical challenges in defining robust similarity metrics, particularly in high-dimensional datasets where determining what makes two individuals "similar" beyond their protected attributes is extremely difficult.

2.2.2 Protected Attributes and Proxy Discrimination

The challenge of achieving fairness is complicated by the phenomenon of proxy discrimination, where seemingly neutral features serve as indirect indicators of protected characteristics. Analysis of real CVs reveals how identity markers are "embedded in broader power relations" and can be revealed through supplementary information, making simple anonymization insufficient for bias prevention [14].

Their qualitative analysis identifies multiple categories of proxy variables:

- Nationality indicators: Location of education, language proficiency patterns, and grammatical structures
- Age proxies: Graduation dates, career timeline patterns, and social media references
- Gender identity markers: Awards mentioning "women," involvement in gender-focused organizations
- Socioeconomic status signals: University prestige, conference attendance records, expensive hobbies, and professional CV formatting quality
- Political affiliation cues: Involvement in labor movements, LGBTQ causes, or environmental activism

This research demonstrates that "relying solely on anonymizing explicit identity markers is a flawed approach to de-biasing hiring," as identities remain detectable through these subtle but systematic patterns in resume content.

2.2.3 Trade-offs Between Fairness and Accuracy

A fundamental challenge in fairness-aware machine learning is the existence of impossibility theorems, which demonstrate that various desirable fairness metrics frequently cannot be simultaneously satisfied, particularly when underlying outcome prevalences differ across protected groups. These theoretical results reveal that "if group membership strongly correlates with the target variable, enforcing exact statistical parity will inevitably compromise predictive accuracy" [14].

Specific conflicts are well-documented in the literature:

Statistical Parity vs. Individual Merit: Achieving demographic parity may require selecting less qualified candidates from underrepresented groups, potentially violating individual fairness principles.

Equalized Odds vs. Demographic Parity: A model can satisfy equalized odds while failing demographic parity requirements.

Calibration vs. Equalized Odds: Predictive calibration (where predicted probabilities reflect true probabilities) is mathematically incompatible with equalized odds in most realistic scenarios.

These trade-offs necessitate careful consideration of context-specific values and priorities. The choice of fairness metric becomes "fundamentally a value judgment" that must account for the specific application domain, potential societal harms, and prevailing legal and ethical frameworks.

2.2.4 Implications for Resume Screening Systems

The theoretical foundations of fairness in machine learning have direct implications for resume screening applications. Proxy discrimination findings suggest that traditional anonymization approaches are insufficient against sophisticated AI systems capable of inferring protected attributes from textual patterns [14, 30]. This creates a fundamental tension: the very features that make resumes informative for job-relevant assessment (writing style, educational background, career trajectory) may simultaneously serve as proxies for protected characteristics.

Furthermore, the impossibility theorems indicate that perfect fairness across all metrics is mathematically unattainable in resume screening contexts where base rates of qualifications may differ across demographic groups. This necessitates explicit choices about which aspects of fairness to prioritize; whether ensuring proportional representation (statistical parity), equal opportunities for qualified candidates (equality of opportunity), or consistent error rates across groups (equalized odds).

The ground truth dependency of most fairness metrics presents additional challenges in hiring contexts, where historical hiring decisions used as training labels may themselves reflect biased human judgments. As noted in the fairness literature, "if the ground truth itself is biased, even technically 'fair' models can perpetuate existing inequalities," making the integrity of training data a paramount concern for fair resume screening systems [31].

2.3 Transfer Learning and Resume Analysis for ICT Classification

Transfer learning has emerged as a critical methodology for addressing the fundamental challenges of limited labeled data and domain specificity in text classification tasks [32]. This section examines the theoretical foundations and practical considerations relevant to applying transfer learning in resume analysis and job categorization contexts.

2.3.1 Domain Adaptation in NLP and Technical Text

Transfer learning in natural language processing leverages knowledge acquired from large, general-purpose datasets to improve performance on domain-specific tasks with limited training data [32]. Investigation of technical logbook datasets reveals insights about domain adaptation effectiveness: "performing transfer learning within a domain provides statistically significant improvements," while cross-domain transfer often degrades performance [33]. Their analysis demonstrates that dataset similarity scores correlate with transfer learning success, suggesting that semantic and linguistic proximity between source and target domains is critical for effective knowledge transfer.

This finding has direct implications for resume classification, where technical domains (such as ICT) possess specialized vocabulary, standardized skill terminologies, and domain-specific career progression patterns that may not transfer effectively from general text corpora.

2.3.2 Resume Classification Approaches and Challenges

The classification of resumes by profession presents unique technical challenges that distinguish it from general text classification tasks. Challenges such as small datasets, lack of standardized resume templates, and privacy concerns hinder the accuracy and effectiveness of existing classification models [34]. The variability in resume formatting, inconsistent terminology usage, and heterogeneous career trajectory descriptions create significant obstacles for automated classification systems. Recent advances using Large Language Models show promising results, with 92% top-1 accuracy on a dataset of 13,389 resumes using BERT-based architectures. However, these results are achieved on relatively large, carefully curated datasets that may not reflect the constraints faced in specialized domain applications or when protected attribute considerations limit data collection [34].

Notably, the task of automated job categorization faces several methodological challenges:

Semantic Ambiguity: Job titles and skill descriptions often use inconsistent terminology across organizations and time periods, making automated categorization prone to misclassification. Skills that are fundamental to one domain may be secondary in another, creating challenges for models trained on broad job market data.
Temporal Evolution: Technical job requirements evolve rapidly, particularly in fields like ICT where new technologies and methodologies continuously emerge. Models trained on historical data may fail to accurately classify contemporary positions that incorporate recently developed skills or technologies.
Custom Mapping Challenges: Classification schemes (such as ICT vs. non-ICT) require subjective mapping decisions that may not align with the nuanced realities of interdisciplinary roles. Positions that combine technical and non-technical responsibilities present particular challenges for automated categorization systems.

2.4 Adversarial Debiasing Techniques

Adversarial debiasing represents a sophisticated approach to bias mitigation that leverages adversarial training principles to learn fair representations. This methodology has emerged as a promising technique for addressing the fundamental trade-offs between predictive accuracy and fairness by forcing models to learn features that are predictive for the main task while being uninformative about protected attributes [8, 35].

2.4.1 Gradient Reversal Layer Theory and Implementation

The Gradient Reversal Layer (GRL) serves as the core mechanism enabling adversarial debiasing by creating a minimax optimization framework between a main predictor and an adversarial discriminator. In the learning not to learn biased representations, the GRL reverses gradients from the bias predictor back to the feature extractor, directly training the feature extractor to produce representations that the bias predictor cannot effectively utilize [8].

The mathematical foundation involves a feature extractor f(x) that learns representations designed to maximize performance on the main task while simultaneously minimizing the adversary's ability to predict protected attributes. The adversarial objective can be expressed as:

$$\min_{\theta_f,\theta_g} \max_{\theta_h} \left(L_c(y, z_c) - \lambda L_B(b, z_B)\right)$$
$$\text{where } z_c = g(f(x)) \quad \text{and} \quad z_B = h(f(x))$$

where L_c represents the classification loss for the main task, L_B represents the bias prediction loss, and λ controls the trade-off between task performance and bias mitigation. The GRL implements this minimax game by reversing the sign of gradients flowing from the bias predictor h back to the feature extractor f during backpropagation [8].

2.4.2 Methodological Considerations

The success of adversarial debiasing depends critically on several methodological factors. The choice of adversary architecture must be sufficiently powerful to detect subtle bias patterns while avoiding overpowering the main predictor during training. The hyperparameter λ requires careful tuning to balance fairness and accuracy objectives, with optimal values typically being task-specific and requiring empirical validation [8, 35].

Training stability represents a persistent challenge in adversarial debiasing frameworks. The competing objectives can lead to optimization difficulties.

Furthermore, adversarial debiasing effectiveness is inherently dependent on the quality and completeness of protected attribute labels during training. The approach assumes that all relevant bias sources are identified and labeled, which may not hold in real-world applications where bias manifests through complex, intersectional, or latent mechanisms not captured in available annotations.

Chapter 3

Methodology and Experimental Design

3.1 Dataset Description

This research employs two complementary datasets to address the challenge of limited data availability. The experimental design leverages transfer learning from a larger, standardized source dataset to improve performance on a smaller target dataset with comprehensive protected attribute labels.

The FINDHR dataset's protected attribute labels enable investigation of bias detection and mitigation, addressing **Research Questions 1, 2 and 3**, while the LiveCareer dataset provides the scale necessary for robust ICT classification modeling.

3.1.1 FINDHR Dataset (Target Dataset)

The primary dataset originates from the FINDHR data donation campaign, collected through Pompeu Fabra University's Institutional Committee for Ethical Review of Projects (CIREP) with full GDPR compliance `https://findhr.eu/`.

The dataset comprises 949 complete CV-survey pairs from 1,139 total submissions collected between June 2023 and May 2024 [36, 37].

Language Distribution: The multilingual nature reflects the European employment environment with four major languages represented: Spanish (69%), English (19%), German, and Catalan.

Professional Categories: The dataset exhibits 50% classified as Professionals across diverse sectors (Science/Engineering, Business/Administration, ICT, Legal, Social Services, Healthcare), 14% Clerical Support Workers, and 12% Service/Sales Workers.

Protected Attribute Survey: Demographic data collection includes age, gender identity, sexual orientation, ethnicity, religion, disability status, national origin, and minority group membership. The survey's three-part structure (CV upload, professional classification, demographic questionnaire) ensures complete attribute pairing with resume content while maintaining participant anonymity.

Table 1 provides an overview of the gender distribution within the FINDHR dataset, specifically highlighting changes observed following the ICT mapping of the present work.

Table 1: FINDHR Dataset: ICT Classification by Gender

Gender	Total	ICT Rate	non-ICT Rate
Women	461 (48.6%)	47.7%	52.3%
Men	438 (46.2%)	46.1%	53.9%
Non-binary	8 (0.8%)	_	_
No answer	42 (4.4%)	_	_

3.1.2 LiveCareer Dataset (Source Dataset)

The transfer learning source dataset comprises 2,400 resumes scraped from LiveCareer.com across 24 predefined job categories. Available under a CC0 (Public Domain) license, this dataset provides the larger, more standardized foundation necessary for effective transfer learning to the FINDHR dataset [38].

Dataset Characteristics: All resumes are in English with consistent formatting due to LiveCareer's standardized templates. The selection criteria (LiveCareer quality score 85-100) ensures high-quality, well-structured resume content that facilitates robust feature extraction during pre-training.

Categorization Structure: The 24 job categories include HR, Designer, Information Technology, Healthcare, Engineering, and Finance, providing comprehensive professional domain coverage.

3.2 Three-Phase Experimental Design

This research employs a three-phase experimental methodology designed to systematically investigate protected attribute detection, transfer learning effectiveness, and adversarial debiasing implementation in resume screening systems. Each phase addresses distinct research questions while building toward a comprehensive understanding of bias mechanisms and mitigation strategies in algorithmic hiring systems.

3.2.1 Phase 1: Protected Attribute Detection in Anonymized Data

Objective and Rationale: This first phase directly addresses **Research Question 1**, investigating the vulnerability in current anonymization practices by testing the hypothesis that modern Large Language Models can successfully detect protected attributes from supposedly anonymized resume text. This investigation phase focuses on anonymization gaps and practical privacy protection against sophisticated AI inference attacks. Our Shifterator results provide essential context for understanding why traditional privacy protection measures may be insufficient in the era of advanced natural language processing systems.

The experimental design specifically targets gender detection capabilities, representing the most commonly protected attributes in employment law with clear patterns of historical discrimination in hiring practices. By establishing baseline detection capabilities on anonymized data, this phase provides the foundation for understanding the necessity of more advanced bias mitigation techniques implemented in subsequent phases.

Data Source and Preprocessing Pipeline

The experimental data originates from the FINDHR dataset's anonymized resume submissions, processed through standardized anonymization procedures that remove explicit demographic identifiers including names, contact information, photographs, and direct references to protected characteristics. The anonymized text retains structural and linguistic patterns that may serve as implicit signals for attribute inference.

The preprocessing pipeline for this Phase implements comprehensive text cleaning procedures designed to standardize input format while preserving linguistically relevant patterns. More detailed information are listed in the Appendix:A

LLM Evaluation Framework

The experimental evaluation leverages HuggingFace's Transformers library accessed through Google's Colab infrastructure, providing access to pre-trained language models and ensuring reproducibility across different model architectures. Three distinct transformer architectures were used for testing:

- Multilingual BERT (mBERT) [39].
- RoBERTa-base [40].
- DistilBERT [41].

The LLMs perform supervised gender classification using anonymized resume text as input and survey-derived gender labels as ground truth. Performance evaluation employs standard binary classification metrics (precision, recall, F1-score) to assess the models' ability to predict candidate gender from supposedly neutral resume content.

Linguistic Pattern Analysis using Shifterator

Shifterator is a computational framework developed for quantifying and interpreting differences between text corpora through information-theoretic measures. This methodology provides interpretable insights into the specific textual features that enable protected attribute detection, revealing the underlying mechanisms of anonymization vulnerabilities [42].

The framework performs Words Shift Analysis, systematically identifying words that contribute most significantly to distributional differences between groups, ranking them by their information content rather than mere frequency differences. This approach reveals vocabulary choices that create the highest "surprise" when transitioning between demographic language models, indicating genuine linguistic markers of group membership.

By identifying specific linguistic features that enable attribute inference, Shifterator analysis informs the development of more sophisticated anonymization techniques and helps understand which textual patterns require targeted intervention in fair machine learning systems.

3.2.2 Phase 2: Two Datasets Transfer Learning and Data Preprocessing

Following the findings of the previous phase, we proceeded to work on the full text from the resumes, transitioning from the anonymized and structured format used in the previous phase, enabling our work to better conform to real-world hiring applications.

Phase 2 establishes an ICT/non-ICT classification architecture, leveraging transfer learning principles and setting the ground for the FINDHR dataset debiasing objective.

Rationale for Transfer Learning: Testing showed that FINDHR dataset's 949 samples proved insufficient for optimal transformer performance. Our LiveCareer-to-

FINDHR approach leverages the LiveCareer dataset's scale for foundational learning before adapting to the target, enabling the acquisition of generalizable resume understanding from standardized data while requiring subsequent adaptation to more diverse resumes.

Data Preprocessing Architecture

For both FINDHR and LiveCareer datasets, the preprocessing pipeline addresses three core challenges: preserving document structure during extraction, maintaining semantic fidelity across languages, and ensuring precise attribute integration.

For full-text extraction, we used PDF processing leveraging PyMuPDF's layout-aware capabilities to extract text while maintaining structural hierarchy.

Regarding the FINDHR dataset, the multilingual processing pipeline employs Facebook's M2M-100 for neural machine translation, converting Spanish, German, and Catalan resumes to English. Manual validation procedures ensured the quality of the translation process.

Protected attribute integration utilizes ID-based matching systems to precisely link resume content with survey-derived demographic information. This framework establishes the foundation for subsequent bias analysis while preserving analytical integrity across the dataset.

Additional information on preprocessing pipeline can be found in the Appendix:B

ICT Classification Methodology

Custom Dictionary Creation and Mapping Logic: The binary ICT classification required systematic mapping from original categorical structures. Through personal research and analysis of contemporary occupational frameworks, custom dictionary-based mappings were developed to transform the heterogeneous categorical labels from both datasets into coherent ICT/non-ICT classifications.

- **LiveCareer Dataset Mapping Strategy:** See Table 6 in the Appendix.

- **FINDHR Dataset ISCO-Based Framework:** See Table 7 in the Appendix.

The mapping logic prioritizes functional ICT dependency over peripheral technology usage, distinguishing between roles where ICT skills constitute core competencies versus supportive tools.

Transfer Learning Architecture

The pre-training phase on the LiveCareer dataset required addressing a fundamental challenge: extracting meaningful textual patterns from severely imbalanced data where ICT positions constituted only 14% of the collection. This imbalance creates a critical bottleneck for transfer learning effectiveness, as standard training approaches would produce models that simply learn to predict the majority class, failing to capture the nuanced linguistic features that distinguish technical professionals from other domains [32].

The architectural solution employs ModernBERT-base [43] as the foundational language model for classification, implementing a carefully orchestrated ensemble of fine-tuning techniques designed to force the model to engage meaningfully with minority class examples during the foundational learning phase. Through strategic hybrid sampling that deliberately rebalances training exposure, focal loss mechanisms that amplify learning signals from difficult-to-classify cases [44], and dynamic weighting systems that ensure minority class patterns contribute proportionally to gradient updates, the pre-training process creates feature representations that are genuinely informative for ICT classification rather than merely reflecting statistical dominance patterns.

More detailed technical specifications for this transfer learning architecture, including hyperparameter configurations and optimization details, are provided in the Appendix:C

3.2.3 Phase 3: Adversarial Debiasing Implementation

This final Phase directly targets **Research Questions 2 and 3** by fine tuning the model obtained in the previous phase on the FINDHR Dataset.

This pipeline aims to achieve meaningful bias reduction without compromising classification utility. The implementation of the adversarial loop comprises an ICT classifier (Main Predictor) a gender prediction component (Adversary) and a a Gradient Reversal Layer (GRL), responsible for the min-max optimization objective

Notably, this methodological choice emerges from an evaluation of available bias mitigation paradigms and their suitability for the resume screening context.

Limitations of Alternative Approaches: Post-hoc fairness interventions, such as threshold adjustment or calibration-based methods, operate on model outputs rather than addressing bias at the representational level. While computationally efficient, these approaches often create artificial decision boundaries that may not generalize across different datasets or employment contexts. Moreover, post-hoc methods frequently exhibit unstable performance when underlying data distributions shift, a common occurrence in dynamic hiring markets where job requirements and candidate pools evolve continuously.

Architectural Advantages of Adversarial Training: Adversarial debiasing addresses these limitations by operating at the feature representation level, forcing the model to learn encodings that are simultaneously informative for job classification and uninformative for gender prediction. This architectural approach ensures that bias mitigation is embedded within the model's fundamental decision-making process rather than applied as an external correction mechanism. The adversarial framework's theoretical foundation in game theory provides mathematical guarantees about the trade-off between task performance and fairness objectives, enabling principled optimization of competing goals. Unlike heuristic approaches that rely on empirical tuning, adversarial training offers a methodology for balancing classification accuracy with bias mitigation through the lambda hyperparameter, providing transparent control over the fairness-utility trade-off.

Gradient Reversal Mechanism and Mathematical Framework: The adversarial min-max design ensures that any feature pattern enabling accurate gender detection actively harms the overall optimization objective, creating direct pressure against discriminatory representations from the very hidden features of the model.

The mathematical foundation centers on the Gradient Reversal Layer (GRL), which implements adversarial ignorance through gradient negation during backpropagation.

Forward Propagation: The GRL acts as an identity function, preserving normal prediction pathways: $GRL(h) = h$
Where h represents the encoded feature representations from the BERT encoder.
Backward Propagation: During backpropagation, the GRL reverses and scales gradients flowing from the gender predictor:
$\frac{\partial L}{\partial h} = -\lambda \times \frac{\partial L_{\text{gender}}}{\partial \text{GRL}(h)}$

This mathematical transformation converts the gender predictor from a collaborative component into an adversarial signal. The feature extractor $\phi(x)$ receives gradients that simultaneously:

– Minimize job classification loss: $\frac{\partial L_{\text{job}}}{\partial \phi(x)}$ (normal gradients)

– Maximize gender classification loss: $-\lambda \times \frac{\partial L_{\text{gender}}}{\partial \phi(x)}$ (reversed gradients)

Minmax Objective: This creates a two-player zero-sum game mathematically expressed as:

$$\min_{\phi} \max_{\theta_{\text{gender}}} \left[L_{\text{job}}(\phi(x), y_{\text{job}}) - \lambda \times L_{\text{gender}}(\theta_{\text{gender}}(\phi(x)), y_{\text{gender}})\right] \tag{3.1}$$

where $\phi(x)$ represents the BERT feature extractor, θ_{gender} the gender classifier parameters, and λ controls the adversarial strength.

Convergence Properties: At equilibrium, the feature extractor learns representations that are:

Highly informative for ICT classification (low L_job) Uninformative for gender prediction (gender accuracy $\approx 50\%$)

The lambda parameter λ governs this trade-off intensity. Higher λ values increase pressure against gender-discriminative features, while $\lambda = 0$ reduces to standard supervised learning without fairness constraints.

Model Configuration

The adversarial debiasing framework follows a four-phase training pipeline, as illustrated in the Figure 1, culminating in a comprehensive evaluation setup that enables bias mitigation assessment.

Step 1: Baseline Model Establishment The debiasing pipeline begins by establishing a robust baseline through fine-tuning the pre-trained ModernBERT model from Phase 2 on the FINDHR dataset, optimizing exclusively for ICT classification performance. This baseline establishes both the performance benchmark and foundation for adversarial development, training without any fairness constraints to establish maximum achievable classification utility.

Step 2: Adversary Pre-training The adversarial architecture is constructed by augmenting the baseline classifier with a dedicated gender prediction component. Critically, this gender predictor undergoes isolated pre-training on the frozen baseline encoder features, establishing a competent adversary capable of detecting gender signals embedded within the learned representations. This pre-training phase is essential as the adversarial mechanism requires a skilled gender predictor to provide meaningful gradient signals for the subsequent min-max optimization process.

Step 3: Lambda Parameter Optimization Following adversary establishment, systematic hyperparameter optimization identifies the optimal trade-off between job classification performance and gender debiasing effectiveness. We perform grid search over lambda values λ=[0.1, 0.5, 1.0, 2.0, 5.0] by running low-epochs adversarial debiasing experiments, determining the gradient reversal strength that best balances competing objectives. During this phase, the baseline ICT predictor attempts to learn job classification while feature representations are penalized through the Gradient Reversal Layer according to the chosen λ whenever the adversary successfully identifies gender information from those features.

Step 4: Adversarial Training with Progressive Scheduling The final training phase employs longer adversarial optimization with progressive lambda scheduling, gradually increasing from zero to the optimal lambda value identified in Step 3. This scheduling approach ensures stable convergence of the min-max objective by allowing the model to initially focus on task learning before progressively introducing bias mitigation pressure [45].

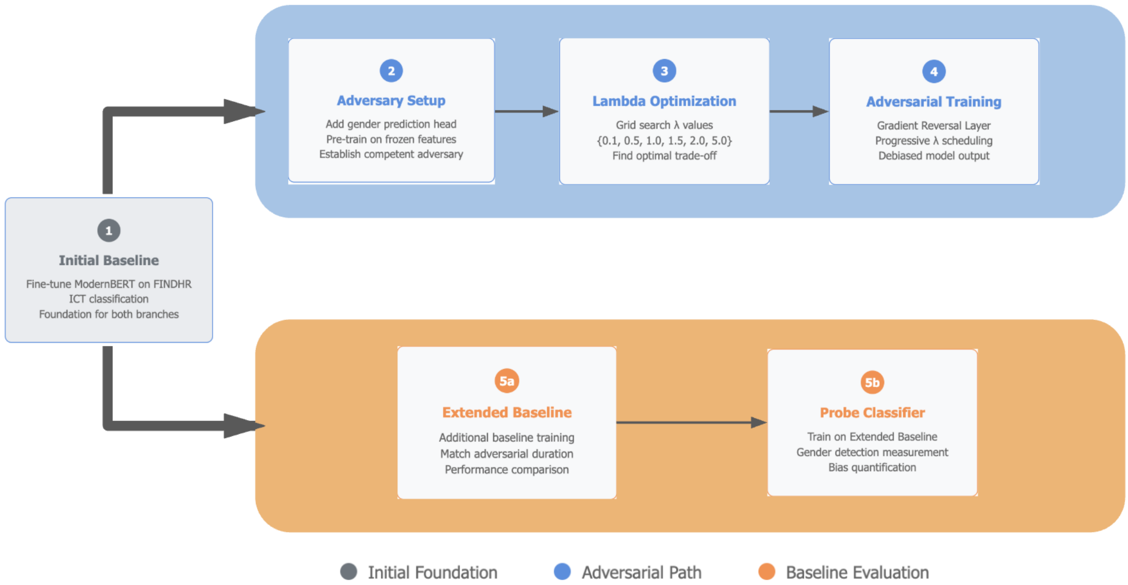

Figure 1: Adversarial Debiasing Pipeline Architecture: Four-step training process (blue) with dual baseline evaluation framework (orange) for bias mitigation assessment.

Evaluation Framework Setup (Steps 5a-5b) To enable the assessment of adversarial debiasing effectiveness, we establish two comparative baselines:

•The Extended Baseline (Step 5a): Additional training matching adversarial duration, optimizing ICT classification performance without fairness constraints to establish a fair job performance comparison without debiasing influence.

•A Probe Classifier (Step 5b): A dedicated neural network trained exclusively on the Extended Baseline's frozen feature representations to predict gender, quantifying the maximum protected attribute's information exploitable from the baseline model's learned embeddings. This establishes the baseline gender bias measurability.

Comparative Assessment Framework This dual baseline architecture enables direct comparison across two critical dimensions:

•ICT Classification Performance: Extended Baseline (no debiasing) versus Adversarial model performance

•Gender Bias Reduction: Probe classifier accuracy on Extended Baseline features versus Adversarial model's built-in gender predictor accuracy on debiased features

This framework ensures that bias mitigation success is measured against realistic baselines while maintaining transparency about the fairness-utility trade-offs inherent in our adversarial debiasing approach.

The adversarial framework's bias mitigation effectiveness stems from its ability to enforce representational independence between task-relevant features and protected attributes. By penalizing any correlation between learned representations and gender information, the methodology moves beyond surface-level bias removal toward fundamental architectural fairness. This approach addresses the proxy discrimination problem identified in Phase 1, where seemingly neutral features serve as indirect indicators of protected characteristics.

This way adversarial debiasing methodology directly addresses **Research Question 3** by providing an approach to evaluating bias reduction effectiveness while maintaining classification performance, establishing clear criteria for success in fair hiring AI applications.

Notably, we define adversarial debiasing effectiveness through three primary criteria:

– Significant reduction in gender predictability from learned representations

– Minimal impact on overall classification performance

– Achievement of improved demographic parity across gender groups.

These criteria provide quantitative benchmarks for evaluating whether adversarial training achieves meaningful fairness improvements without compromising operational utility.

Additional details of the adversarial pipeline architecture, including hyperparameter configurations and optimization details, are provided in AppendixD

3.3 Evaluation Framework

This section establishes the evaluation methodology for assessing both classification performance and bias mitigation effectiveness across all three phases.

Performance Metrics Classification performance evaluation centers primarily on F1-score, chosen specifically for its resistance to overfitting patterns that emerged as a persistent factor throughout this investigation.
This choice reflects practical considerations: precision and recall individually proved insufficient for capturing model behavior in our classification context, where models frequently learned to exploit statistical patterns rather than semantic content.

Bias Measurement Bias evaluation is performed in the Adversarial Debiasing Phase, and employs multiple complementary metrics that capture different aspects of fairness and potential discrimination patterns.
Primary evaluation focuses on gender prediction accuracy from learned features representations, where successful debiasing should reduce accuracy to near-random levels (50%), indicating that gender information has been successfully removed from the feature space.
For effective comparison, gender predictability assessment employs a probe classifier methodology, where a dedicated neural network is trained to predict gender from the frozen feature representations extracted by the extended baseline model. This way we quantify the maximum gender bias that could be exploited by accessing to the model's internal representations.

Next, The demographic parity difference quantifies disparate impact through:

$|P(\text{ICT}|\text{Male}) - P(\text{ICT}|\text{Female})|$

Measuring whether classification probabilities vary systematically across gender groups. Values approaching zero indicate perfect equitable treatment regardless of gender.

Finally, Gender Prediction Rates measure the proportion of positive ICT classifications within each demographic group, formally defined as:
Female Prediction Rate $= P(\hat{y} = 1|g = \text{female}) = \frac{\sum_{i:g_i=\text{female}} \mathbf{1}[\hat{y}_i=1]}{\sum_{i:g_i=\text{female}} 1}$
Male Prediction Rate $= P(\hat{y} = 1|g = \text{male}) = \frac{\sum_{i:g_i=\text{male}} \mathbf{1}[\hat{y}_i=1]}{\sum_{i:g_i=\text{male}} 1}$
where $\hat{y}_i \in \{0, 1\}$ represents the predicted ICT classification for individual i, g_i denotes the gender attribute, and $\mathbf{1}[\cdot]$ is the indicator function.

Chapter 4

Results and Analysis

This chapter covers the results obtained from the implementation of our 3-Phase pipeline described in the Methodology. First, we analyse the LLMs F1 score on the anonymised gender classification together with the Shifterator results. Subsequently, we move to the performances of the Transfer Learning and Adversarial model, listing numerical results in ICT Prediction accuracy and Debiasing performances.

4.1 Protected Attribute Detection Results

Despite anonymization efforts that removed explicit demographic identifiers, modern language models successfully inferred gender information from supposedly neutral resume text with concerning accuracy. This finding challenges the fundamental assumption that removing names and contact details provides adequate privacy protection in automated hiring systems.

Baseline LLM Performance The results of the three transformer models (mBERT, RoBERTa-base, DistilBERT) demonstrate varying but significant detection capabilities, as shown in the table 2.

Table 2: Results on anonymized resumes gender prediction

Model	Test Accuracy	F1 Score	Precision	Recall
mBERT	0.5181	0.3536	0.2684	0.5181
RoBERTa-base	0.6145	0.6146	0.6154	0.6145
DistilBERT	**0.6386**	**0.6376**	0.6434	0.6386

The most effective model, DistilBERT, achieved 63.86% accuracy in gender prediction, substantially above random chance (50%). This performance indicates that anonymization alone is insufficient to prevent protected attribute inference, as sophisticated language models can exploit subtle linguistic patterns and contextual cues embedded within resume content.

4.1.1 Linguistic Pattern Analysis

The Shifterator analysis revealed remarkable differences in vocabulary usage between male and female CVs, providing insight into some of the mechanisms that enable successful gender detection. The figure 2 illustrates the most distinctive words contributing to gender-based classification.

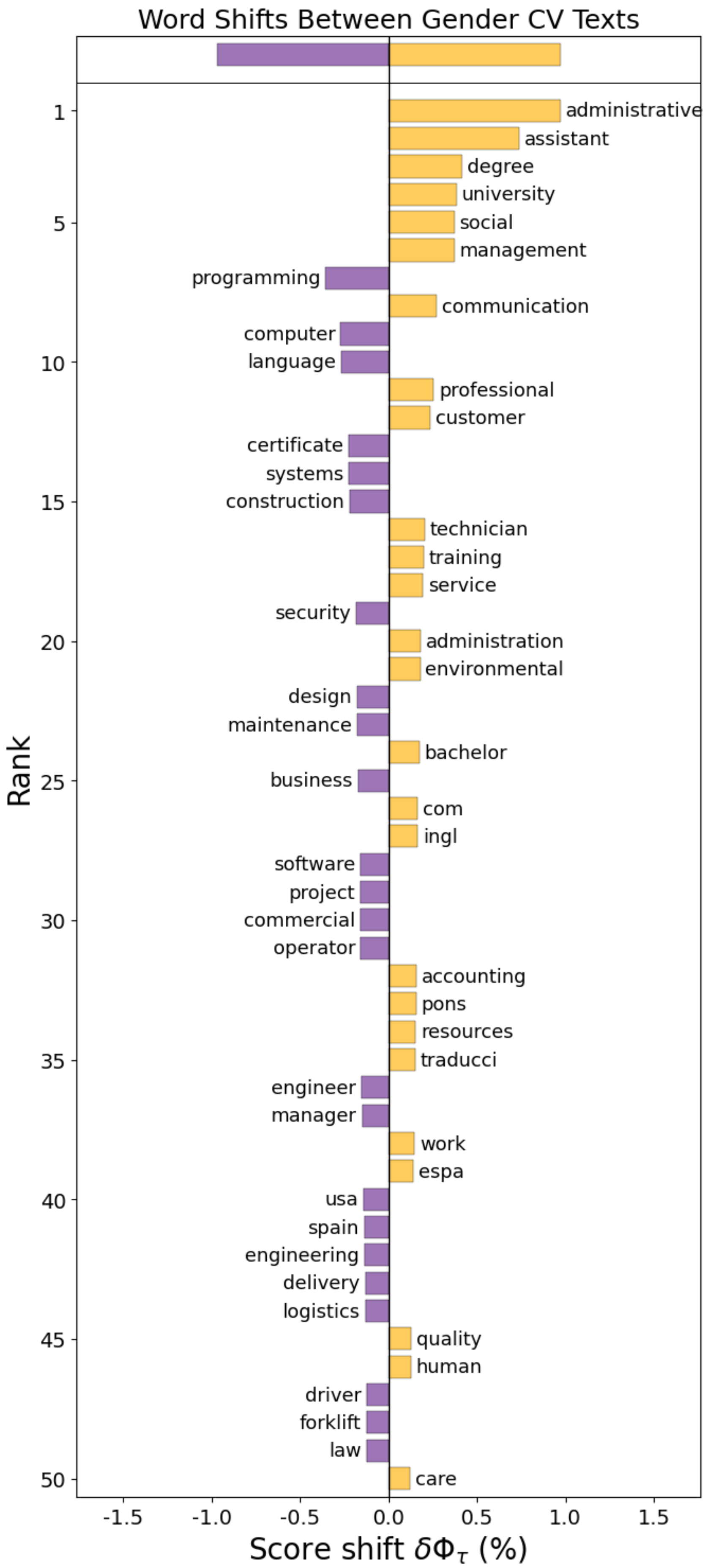

Figure 2: Shifterator: word shift graph. purple = Male | yellow = Female

Figure 2 shows clear patterns in professional self-presentation, identifying terms that are used with notably different frequencies between Male and Female Resumes.

These linguistic markers extend beyond obvious gender indicators and encompass subtle differences in the way professional experiences are described, the usage of technical terminology, and career positioning strategies. The presence of these patterns suggests that even careful anonymization cannot eliminate the textual signals that enable algorithmic bias detection.

Moreover, we used Shifterator for Categorical Content Analysis, going beyond individual word differences and analysing content categories.

Figure 3 illustrates the percentage of vocabulary devoted to different professional categories across gender groups.

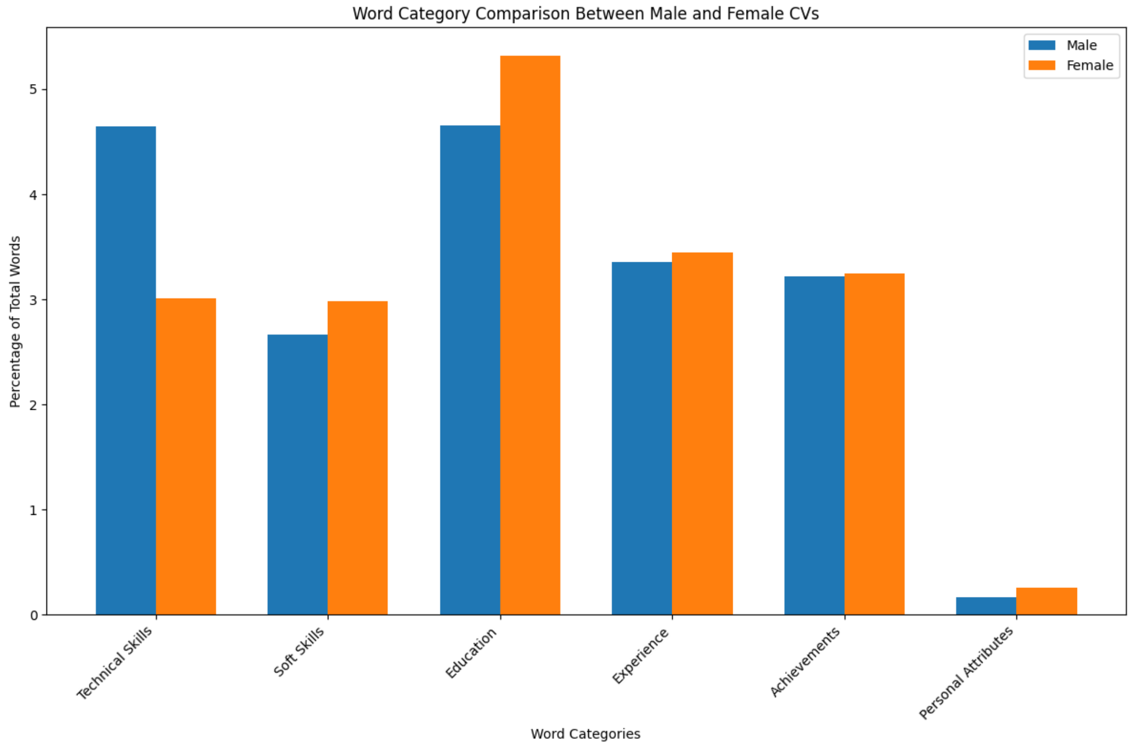

Figure 3: Shifterator: Categorical Content Analysis

Figure 3 reveals distinct patterns in how male and female candidates structure their professional narratives. Three primary areas can be observed:

Technical Skills Emphasis: Male CVs dedicate 4.65% of their vocabulary to technical skills terminology, compared to 3.01% for female CVs. This disparity reflects both occupational segregation and potential differences in how technical competencies are presented professionally.

Educational Focus: Female CVs allocate 5.32% of content to educational credentials and achievements, exceeding male CVs by 0.66 percentage points. This pattern suggests different strategies for establishing professional credibility, with women candidates potentially focusing more on formal qualifications.

Soft Skills Integration: Female CVs incorporate soft skills terminology 0.31 percentage points more frequently than male CVs (2.98% vs. 2.67%), indicating differences in self-presentation approaches and professional positioning strategies.

4.2 Adversarial Debiasing Evaluation

Having established that anonymized resumes remain vulnerable to gender inference through subtle linguistic patterns, we now turn to evaluating mitigation strategies.

The following section examines whether adversarial training had successfully reduced these biases while maintaining classification performance.

4.2.1 Transfer Learning Results on LiveCareer Dataset

The transfer learning approach successfully established robust ICT classification capabilities on the larger LiveCareer dataset before adaptation to the FINDHR dataset for bias analysis and adversarial training.

However, this process revealed significant challenges in maintaining performance across different data distributions.

The ModernBERT-based architecture demonstrated strong results on the LiveCareer dataset 3, effectively learning to distinguish ICT from non-ICT positions despite severe class imbalance (14% ICT positions). The hybrid sampling methodology and focal loss implementation successfully addressed the minority class learning challenge.

Table 3: Test Set Classification Report (using optimal threshold)

Class	**Precision**	**Recall**	**F1-Score**	**Support**
non-ICT	0.98	0.92	0.95	426
ICT	0.65	0.87	0.75	71
weighted avg	0.93	0.92	0.92	497

The model achieved an overall accuracy of 92% (calculated from the total support of 497).

For the majority non-ICT class, the model exhibited high precision (0.98), indicating that when it predicted a job as non-ICT, it was correct 98% of the time. The recall for this class was also strong at 0.92, meaning that 92% of actual non-ICT jobs were correctly identified.

Crucially, despite the class imbalance, the model achieved a recall of 0.87, signifying that 87% of all actual ICT positions were correctly identified by the model. While the precision for the ICT class was lower at 0.65, meaning 65% of instances predicted as ICT were indeed ICT, this still represents a substantial capability to identify the minority class.

The weighted average F1-score of 0.92 further confirms the model's overall strong and balanced performance across both classes, taking into account their respective proportions in the dataset.

To further illustrate the model's predictive behavior, we present the confusion matrix for the test set in Figure 4.

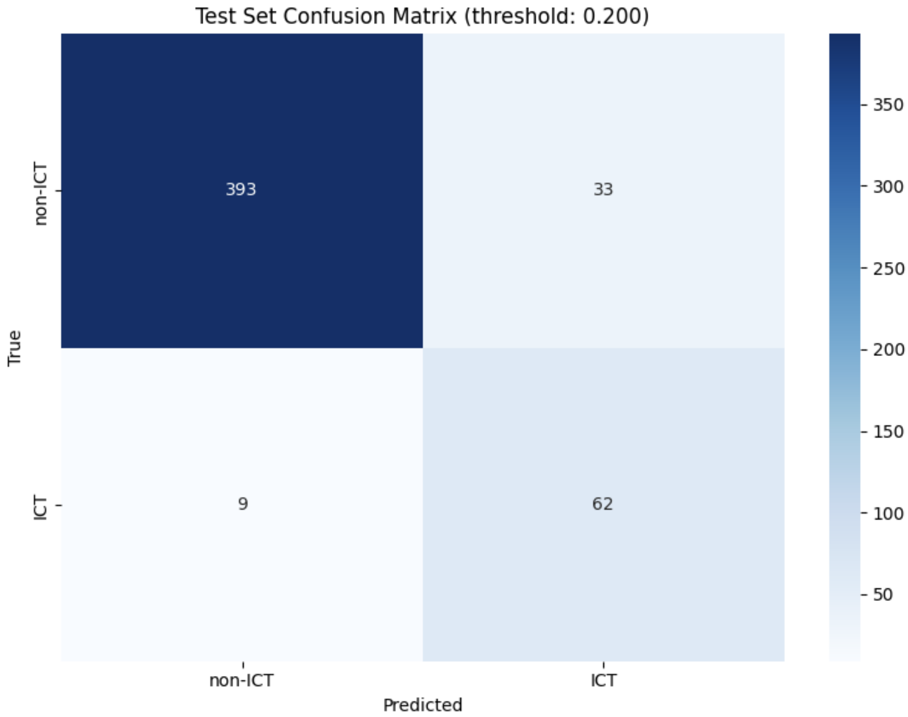

Figure 4: Confusion Matrix LiveCareer performance

It can be observed that out of 426 true non-ICT instances, 393 were correctly classified (True Negatives), while 33 were incorrectly classified as ICT (False Positives).

For the ICT class, out of 71 true instances, 62 were correctly identified (True Positives), and only 9 were missed and incorrectly classified as non-ICT (False Negatives).

4.2.2 Transfer to FINDHR Dataset

The transition from LiveCareer to FINDHR revealed significant domain adaptation challenges, with performance dropping significantly despite the transfer learning approach.

The FINDHR fine-tuning phase achieved a best overall F1 score of 0.67 (from the 0.92 on LiveCareer data).

To illustrate the model's learning trajectory and adaptation challenges during the fine-tuning process on FINDHR, the training and validation accuracy curves for each of the three folds are presented in Figure 5:

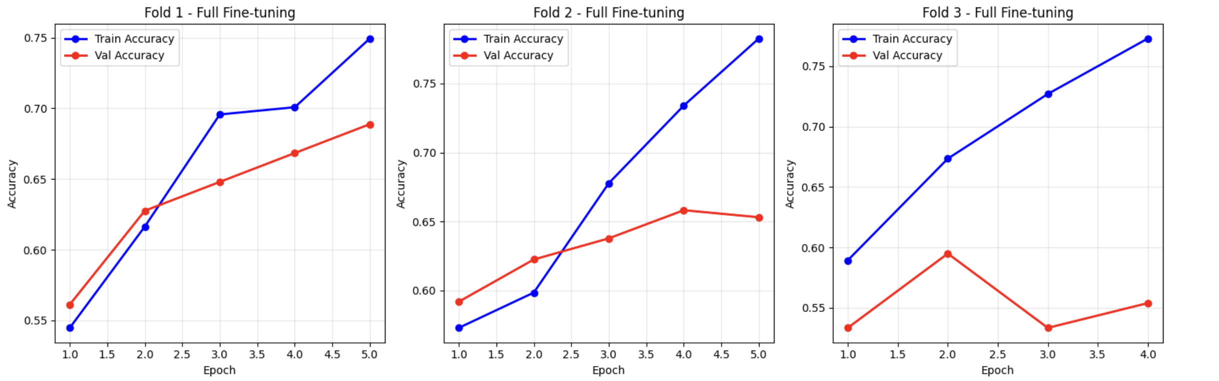

Figure 5: CrossValidation learning curves for full fine-tuning phase

While the training accuracy generally shows an upward trend, indicating the model's ability to learn from the training data, the validation accuracy exhibits variability and plateaus or even declines in some folds (Fold 3 after epoch 2), suggesting potential overfitting to the specific training data and limited generalization to unseen data from the new domain.

Nonetheless, the graphs 5 demostrate correct learning patterns, with the best F1-score of 0.67 for the FINDHR dataset reflecting acceptable learning outcomes.

4.2.3 Pre-training of Gender Predictor

Next, the gender classifier component underwent isolated pre-training to ensure the Adversary is capable of detecting gender signals in the encoded representations before proceeding with the gender-features removal.

This pre-training process demonstrated clear learning progression, with gender detection accuracy improving from an initial 52.76% (random chance) to a final 67.6%, proving the model developed understanding of gender information.

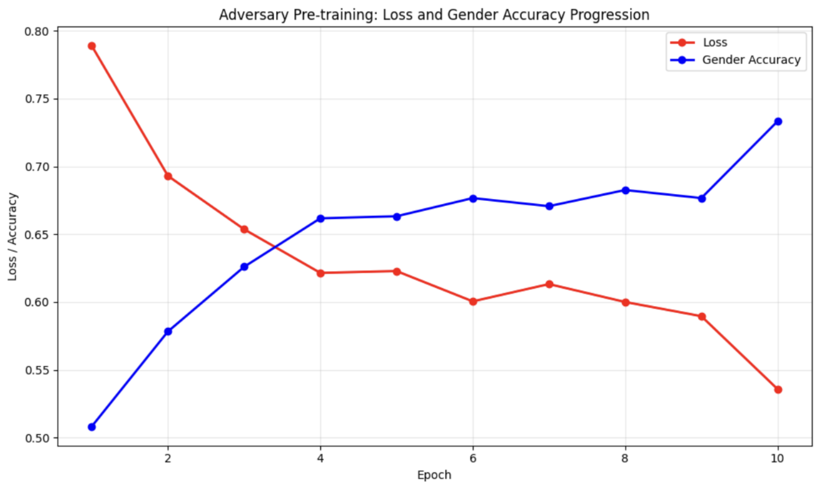

Figure 6: Adversary pre-training progression

The training curves 6 showed consistent improvement, with the gender classifier achieving stable performances by epoch 4, indicating successful learning of gender-discriminative features from the frozen BERT representations. The final accuracy confirms that the gender classifier could reliably detect protected attributes embedded within the job classification features.

4.2.4 Lambda Parameter Optimization

Following adversary establishment, a systematic grid search was conducted to identify the optimal lambda (λ) parameter to best balance ICT classification performance with gender debiasing effectiveness.

Table 4: Lambda Parameter Optimization Results

Lambda (λ)	Job F1 Score	Gender Accuracy
$\lambda = 0.1$	0.5455	0.6190
$\lambda = 0.5$	0.5546	0.6429
$\lambda = 1.0$	0.5476	0.5357
$\lambda = 2.0$	**0.5870**	**0.4524**
$\lambda = 5.0$	0.5359	0.4643

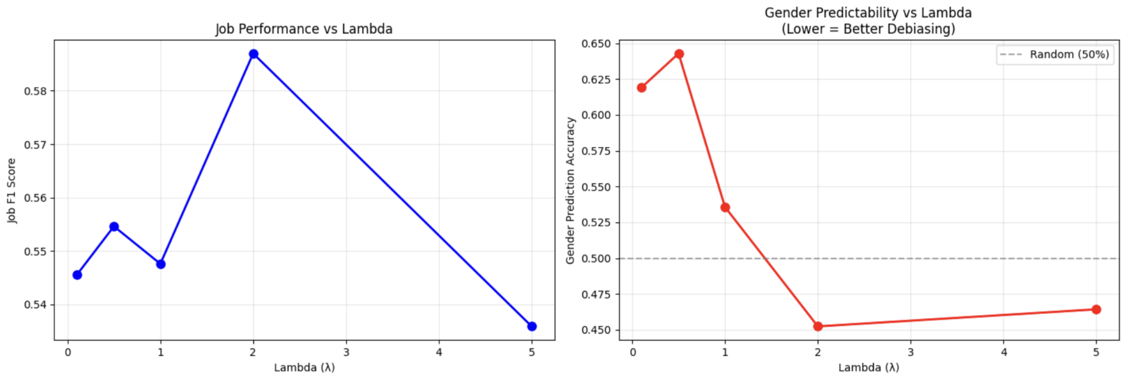

Figure 7: ICT-classification vs Debiasing tradeoffs

The lambda parameter controls the strength of the gradient reversal mechanism, determining how aggressively the model is penalized for producing gender-informative features.

The results 4 demonstrate a clear trade-off pattern between job classification performance and gender debiasing effectiveness. Lower lambda values (λ=0.1, λ=0.5) maintained moderate job classification accuracy but failed to adequately suppress gender detectability, with the adversary achieving up to 64.29% gender prediction accuracy.

Conversely, higher lambda values increasingly penalized gender-discriminative features, with λ=5.0 achieving optimal debiasing performance by reducing gender accuracy to random chance levels.

On the other hand, as lambda values increased, systematic bias mitigation emerged. The progression shows λ=1.0 successfully reduced gender accuracy to 53.57%, approaching random chance levels, while λ=2.0 achieved the optimal balance by further reducing gender detectability to 45.24% while simultaneously achieving the highest job classification performance (F1=0.5870).

Higher lambda values λ=5.0 continued bias mitigation with an even better gender accuracy of 46.43%, but at significant cost to classification utility, with F1-score dropping to 0.5359, indicating a threshold of over-regularization that compromises operational performance.

The lambda value of 2.0 was selected as optimal, representing the ideal configuration that maximizes both job classification capability and gender debiasing effectiveness. This setting achieved the strongest ICT prediction performance while maintaining gender prediction accuracy well below random chance, indicating successful bias mitigation without performance degradation.

4.2.5 Full adversarial Training

Following lambda parameter optimization, the final adversarial training phase represents the culmination of our bias mitigation approach. Using the same training setup as lambda selection and the optimal parameter value (λ=2.0) identified in the previous phase, we conducted extensive adversarial training with progressive lambda scheduling to ensure stable convergence of the competing objectives.

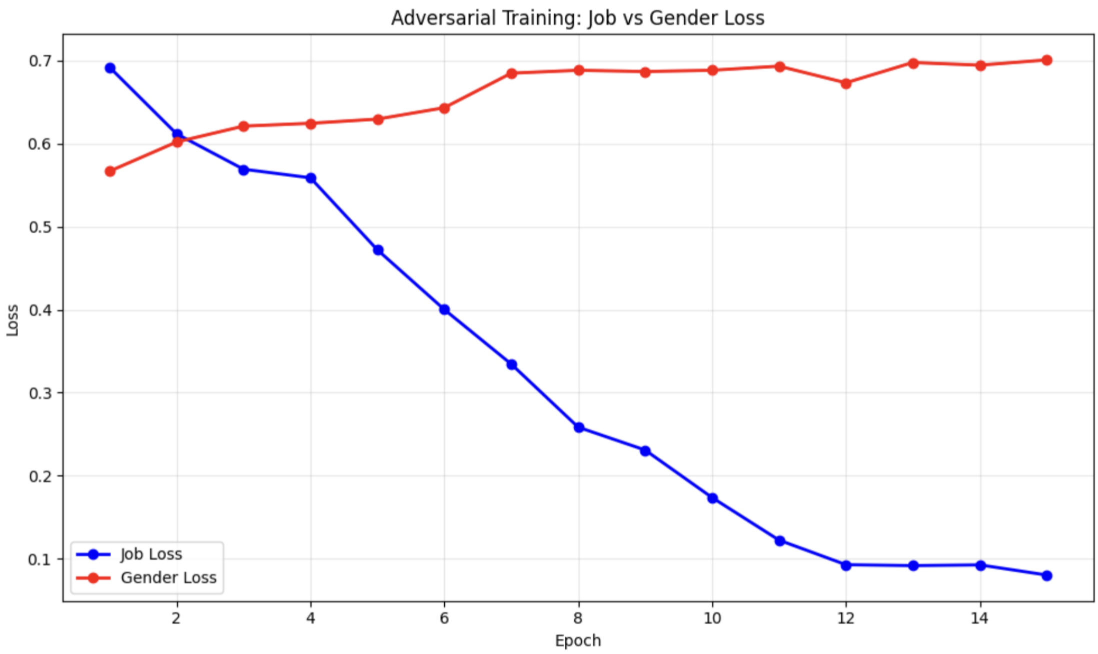

Figure 8: Adversarial Training Dynamics: ICT classifier Loss vs Gender Loss Over Training Epochs

The graph 8 illustrates the adversarial training dynamics, demonstrating the successful implementation of the min-max optimization objective. The job loss (blue line) exhibits consistent downward progression, indicating continuous improvement in ICT classification capability throughout training. Simultaneously, the gender loss (red line) increases from 0.56 to 0.70, reflecting the adversary's growing inability to predict gender from the learned representations and almost reversing to before training performances, thus achieving precisely the desired outcome for successful debiasing.

This divergent loss pattern confirms the effectiveness of the gradient reversal mechanism: as the model's extracted features become more proficient at job classification, they simultaneously becomes less informative about protected attributes. The smooth convergence without oscillation validates our progressive scheduling approach and demonstrates stable adversarial optimization.

The performance on the test set shows that the adversarial model achieved remarkable reduction in bias while maintaining operational utility. Gender predictability was reduced to 51.19%, effectively approaching random chance (50%). This near-random performance indicates successful removal of gender-discriminative patterns from the learned feature representations. Critically, this bias mitigation was achieved without sacrificing classification performance. The model maintained a job accuracy of 63.69%, demonstrating that fairness and utility can be simultaneously optimized through adversarial training architectures.

4.2.6 Baseline and Probe gender predictor Training

To establish rigorous comparative benchmarks for adversarial debiasing assessment, we integrated extended training of the Baseline model alongside bias measurement through probe classifier methodology.

The baseline model underwent additional 15 epochs of fine-tuning, matching the training steps of the adversarial pipeline and optimizing ICT classification performance without any fairness constraints. This extended baseline achieved strong classification performance with a job accuracy of 64.88% and an overall F1-score of 0.6443, establishing the upper bound for classification utility without bias mitigation interventions.

However, this performance came at the cost of embedded bias, as evidenced by subsequent probe analysis. Gender bias measurement employed a dedicated probe classifier trained exclusively on the extended baseline model's frozen feature representations. This probe achieved 63.10% gender prediction accuracy, almost matching our initial LLMs investigation and demonstrating substantial gender information embedded within the supposedly task-focused learned representations.

These baseline results establish both the performance ceiling and the bias floor against which adversarial debiasing effectiveness can be rigorously evaluated, providing essential context for understanding the fairness-utility trade-offs achieved through our architectural interventions.

4.2.7 Final Models Comparison and Bias Reduction Performance

Our comparative evaluation between baseline and adversarial models reveals the effectiveness of Adversarial Debiasing in achieving meaningful bias reduction while preserving operational utility.

Table 5 presents the comprehensive performance comparison across all critical metrics.

Table 5: Performance comparison between baseline and adversarial models

Model	**Job Accuracy**	**F1 score**	**Gender Accuracy**	**Demo Parity**
Baseline	0.6488	0.6443	0.6310	0.0147
Adversarial	0.6369	0.6287	0.5119	0.0060
Difference	**-1.19%**	**-1.56%**	**-18.9%**	**-59%**

With respect to classification performance trade-off, the adversarial model achieved comparable classification performance to the baseline, with job accuracy declining modestly from 64.88% to 63.69% (1.19 percentage point decrease) and overall F1-score from 0.6443 to 0.6287. This minimal performance cost is encouraging and demonstrates that fairness constraints have minimal impact on ICT classification performances.

On the other hand, bias mitigation proved very effective. The adversarial training produced substantial bias reduction across all fairness metrics. Gender predictability decreased from 63.10% to 51.19%, representing an 18.9% relative reduction in exploitable gender information. This reduction brings gender detection capability effectively to random chance levels (50%), indicating successful architectural debiasing. Demographic parity improvements were equally compelling, with the disparity difference reduced from 0.0147 (Female Prediction Rate: 0.4043 - Male Prediction Rate: 0.4189) to 0.0060 (Female Prediction Rate: 0.3723 - Male Prediction Rate: 0.3784) a 59% reduction in differential treatment across gender groups. This improvement demonstrates that adversarial training addresses not only individual fairness (reduced gender detectability) but also group fairness (equalized treatment rates).

Chapter 5

Discussion

5.1 Key Findings and Contributions

Protected Attribute Detectability: This research directly answers **Research Question 1** by establishing that anonymization provides inadequate protection against modern AI inference attacks. With our findings using DistilBERT (63.86% gender detection accuracy on anonymized resumes) we prove that removing explicit identifiers fails to eliminate protected attribute signals embedded in professional language patterns.

Our investigation with Shifterator targets these signals, providing transparency on some of the mechanisms enabling this detection. Male resumes allocated 4.65% of vocabulary to technical terminology compared to 3.01% for female resumes. On the other hand, Female resumes dedicated 5.32% of content to educational credentials, exceeding male resumes by 0.66 percentage points. Additionally, professional terminology and personal presentation also systematically differ across demographic groups.

Thus, linguistic patterns persist despite anonymization, creating algorithmic vulnerabilities in current privacy protection strategies. These differences likely reflect broader structural inequalities in education, career development, and professional networking that are detected when processed by sophisticated AI systems.

This findings challenge fundamental assumptions underlying GDPR compliance in algorithmic hiring. Organizations implementing standard anonymization procedures may unknowingly process protected attributes through algorithmic inference, potentially violating data protection regulations while believing themselves compliant.

Therefore, the present research exposes critical gaps between regulatory intent and technical reality. Current anonymization standards focus on explicit identifier removal while ignoring implicit signals, this vulnerability demands innovative solutions.

Gender Bias in ICT Classification: Next, we directly address **Research Question 2** by quantifying gender bias embedded within our classification scope of ICT versus non-ICT positions. Our investigation reveals that while the FINDHR dataset exhibits balanced gender representation in ICT positions (47.7% women, 46.1% men), this homogeneity masks underlying bias.

In our analysis we show that ICT features extraction still retains relevant gender information, resulting in 67% detection accuracy during adversary pre-training and 64.9% according to final testing on the Extended Baseline encoder.

This finding is particularly significant because hidden features extracted by neural networks fundamentally represent the patterns the model has learned are useful or strongly correlated with the prediction task. In machine learning, feature learning operates through gradient descent optimization, where the model systematically identifies and amplifies input patterns that minimize prediction loss. When training for ICT classification, the model discovers that certain linguistic patterns, educational backgrounds, and career trajectory markers effectively distinguish ICT from non-ICT roles.

However, these apparently job-relevant features inadvertently encode gender information due to historical biases and societal inequalities embedded within professional development pathways. The model learns that specific technical terminology usage, educational institution types, and career progression patterns predict ICT roles, yet these same patterns systematically correlate with gender.

Thus, the 64.9% gender detection accuracy from ICT-trained features proves that seemingly neutral job classification criteria are contaminated with protected attribute signals, validates the proxy discrimination hypothesis.

Adversarial Debiasing Implementation Success: This research concludes by answering **Research Question 3**, demonstrating in our implementation that adversarial training can achieve optimal bias reduction while maintaining operational utility.

Using gradient reversal architecture, we prove that architectural fairness interventions is a valuable tool compared to post-hoc bias correction approaches. With our implementation gender prediction accuracy decreased from 63.10% to 51.19%, representing an 18.9% reduction in demographic signal detectability, while ICT classification performance declined minimally, retaining 97.6% of baseline F1 score (0.6287 versus 0.6443). Most significantly, demographic parity improved by 59%, reducing disparate impact from 0.0147 to 0.006, indicating substantially more equitable treatment across gender groups.

Moreover, our investigation through lambda parameter optimization proves useful to balance protected attribute debiasing with performance loss, preventing some of the results degradation that typically accompanies traditional fairness interventions. Thus, by embedding fairness constraints within feature learning processes, we address proxy discrimination at its representational source. This validates that adversarial debiasing represents a principled approach to achieving fairness with minimal sacrifice of predictive performance, offering organizations transparent control over fairness-utility trade-offs through the lambda hyperparameter.

5.2 Limitations and Challenges

Notably, we acknowledge the following main limitations of the present study.

Datasets Challenges The FINDHR dataset's 949 samples proved insufficient for optimal language model performance, contributing to suboptimal ICT classification

F1 score. Cross-validation curves indicated overfitting patterns that larger datasets would likely mitigate. Moreover the demographic balance in ICT positions (47.7% women, 46.1% men) significantly differs from industry norms where women typically represent 17-20%. This shift may have reduced the detectability of systematic gender bias patterns that would be more pronounced in datasets reflecting actual market imbalances.

Additionally, the participant-driven anonymization process, while preserving authenticity, introduces variability in information removal that could affect algorithmic consistency. The dataset's temporal scope (11 months) captures a specific labor market period, potentially missing broader employment variations or longer-term economic trends.

On the other hand, some limitations should be noted also in the LiveCareer Dataset. A source bias exists as data originates solely from LiveCareer.com, potentially not capturing the full diversity of resume styles. Furthermore, while the high-score filter ensures quality, it may skew the dataset towards more idealized resume examples.

Scope and Methodological Limitations This research explicitly constrains its scope to binary gender classification, excluding non-binary gender identities. This limitation extends beyond simple omission, binary classification frameworks actively reinforce gender binaries by treating non-conforming identities as "out-of-distribution" cases.

Such systems may inadvertently strengthen discrimination against gender-diverse individuals by algorithmically encoding the assumption that only binary categories represent valid gender expressions.

Additionally, the exclusive focus on gender bias ignores other protected attributes including race, age, disability status, sexual orientation and many more. Employment discrimination manifests across numerous demographic dimensions simultaneously, yet this research provides no framework for multi-attribute bias detection or mitigation.

Furthermore, the binary ICT versus non-ICT classification oversimplifies contemporary job market realities. This methodological choice enables experimental control but limits practical applicability to complex organizational structures.

The singular focus of this study on adversarial training excludes alternative bias mitigation approaches. This constraint ignores comparative evaluation of competing methodologies and may overlook more effective approaches for specific deployment contexts.

Performance Trade-offs in Debiased Systems Adversarial training necessarily reduced task performance compared to biased baselines. The optimal adversarial model achieved F1=0.6287 versus the baseline's F1=0.6443, representing a meaningful utility cost for bias mitigation. Organizations must explicitly weigh these trade-offs against legal, ethical, and reputational risks of biased systems.

In the case of this study, this can be controlled using the λ parameter.

Evaluation Metrics Limitations We acknowledge that computational evaluation often results insufficient in capturing real world dynamics, like complex discrimination patterns in real hiring contexts. Demographic parity and equalized odds metrics provide mathematical precision but may not align with legal or ethical conceptions of fair treatment in employment decisions.

5.3 Implications and Future Work

With this research we directly informs AI Act Article 10(5) by demonstrating practical approaches to bias monitoring and detection in high-risk AI systems.

Our findings challenge current GDPR anonymization standards by revealing vulnerabilities to inference attacks. Regulatory frameworks require updating to address sophisticated AI capabilities that current privacy protection measures inadequately anticipate.

Industry Applications Our adversarial training methodology offers immediate practical value for HR technology validation. A similar pipeline could become industry standard practice for testing hiring AI systems before deployment, providing robust approaches to bias detection and mitigation.

Continuous monitoring systems represent logical extensions of this work. Rather than one-time bias detection, adversarial training could enable ongoing bias mitigation that adapts to changing hiring market conditions and evolving demographic patterns in applicant pools.

Finally, by tuning the lambda parameter, organizations could dynamically regulate their fairness-utility trade-offs in addition to introducing transparency in their algorithmic decision-making processes.

Technical Extensions Multi-attribute fairness represents the most critical technical advancement required. Extending beyond binary gender classification to intersectional bias detection would address real-world discrimination patterns that single-attribute approaches cannot capture.

In addition, legal-technical partnerships represent essential collaborations for developing compliance-by-design systems. Technical bias mitigation requires legal expertise to ensure regulatory alignment and practical enforceability in employment contexts.

Similarly, sociolinguistic collaboration could advance understanding of the linguistic bias patterns this research identified.

Finally, industry-academia consortiums for sharing anonymized hiring data could accelerate bias detection research while protecting organizational confidentiality. Collaborative datasets would enable more robust methodological development than individual organizations can achieve independently.

5.4 Conclusions

This research demonstrates that fair AI hiring represents both a technical challenge and a practical necessity. Our findings show that adversarial debiasing achieved an 18.9% reduction in gender bias (from 63.10% to 51.19% detectability) while retaining 97.6% of classification performance, proving that architectural fairness interventions can deliver substantial bias mitigation with minimal utility costs.

However, bias mitigation requires sustained interdisciplinary collaboration rather than purely technical solutions. Dataset constraints, evaluation challenges, and generalizability concerns demonstrate that algorithmic fairness depends on broader social and regulatory contexts that individual research projects cannot fully address.

The future of AI hiring lies not in choosing between operational utility and ethical compliance, but in engineering systems that achieve both through intelligent trade-off optimization. This research provides foundational evidence that such systems are technically feasible and practically necessary for organizations to navigate the intersection of AI capabilities and regulatory requirements.

The urgency of this work increases with AI proliferation in hiring contexts. As automated systems become more sophisticated and widely adopted, the window for implementing effective bias detection and mitigation strategies narrows. In response, the architectural approach to fairness demonstrated in this study offers a path forward that balances innovation with accountability, efficiency with equity, and technical possibility with social responsibility.

List of Figures

List of Tables

Bibliography

[1] Blessing, M. Ai-powered resume screening: Benefits and challenges (2025).

[2] Automated vs. Manual Remote Candidate Screening — potis.ai. https://www.potis.ai/blog/automated-vs-manual-remote-candidate-screening. [Accessed 30-05-2025].

[3] 90 Hiring Statistics: Challenges, Strategies, and Trends — flair.hr. https://flair.hr/en/blog/hiring-statistics/. [Accessed 30-05-2025].

[4] Tarun.B, Fasidh, M. & Nithya, M. S. Job screen AI – automated resume screening system. *Int Res J Adv Engg Mgt* **3**, 882–885 (2025).

[5] Zhu, Y. Ai and employment discrimination: Aihr's algorithmic bias. *Applied and Computational Engineering* **118**, 89–94 (2025).

[6] Soleimani, M., Intezari, A., Arrowsmith, J., Pauleen, D. J. & Taskin, N. Reducing AI bias in recruitment and selection: an integrative grounded approach. *Int. J. Hum. Resour. Manag.* 1–36 (2025).

[7] Rafi, M. When AI plays favourites: How algorithmic bias shapes the hiring process — theconversation.com. https://theconversation.com/when-ai-plays-favourites-how-algorithmic-bias-shapes-the-hiring-process-239471 [Accessed 30-06-2025].

[8] Kim, B., Kim, H., Kim, K., Kim, S. & Kim, J. Learning not to learn: Training deep neural networks with biased data (2018). 1812.10352.

[9] Heruy Mengistu, P. Automating Resume Classification — heruymengistu. https://medium.com/@heruymengistu/automating-resume-classification-e449bff2fadf. [Accessed 02-07-2025].

[10] Sharan, G. HR technology adoption: Trends, growth, and impact — anz.peoplemattersglobal.com. https://anz.peoplemattersglobal.com/article/hr-technology/hr-technology-adoption-trends-growth-and-impact-45727. [Accessed 30-05-2025].

[11] Coresignal. What is AI Resume Screening and Do You Need It? — coresignal.com. https://coresignal.com/blog/ai-resume-screening/. [Accessed 02-07-2025].

[12] Chang, X. Gender bias in hiring: An analysis of the impact of amazon's recruiting algorithm. *Advances in Economics, Management and Political Sciences* **23**, 134–140 (2023).

[13] Filipsson, F. Amazon AI Hiring Tool: A Case Study in Algorithmic Bias — redresscompliance.com. https://redresscompliance.com/amazon-ai-hiring-tool-a-case-study-in-algorithmic-bias/#:~:text=However,%20Amazon's%20experiment%20with%20an,Amazon%20to%20scrap%20the%20project. (2025). [Accessed 30-05-2025].

[14] Vinod Bhatia, K., Capasso, M., Arora, P., Castillo, C. & Saldivar, J. Proxy discrimination risks in hiring: A qualitative analysis of a set of real CVs (2025).

[15] General Data Protection Regulation (GDPR) – Legal Text — gdpr-info.eu. https://gdpr-info.eu/. [Accessed 02-07-2025].

[16] irjet.net. https://www.irjet.net/archives/V12/i4/IRJET-V12I4268.pdf. [Accessed 02-07-2025].

[17] Women in the ICT sector — eige.europa.eu. https://eige.europa.eu/publications-resources/toolkits-guides/work-life-balance/women-in-ict?language_content_entity=en. [Accessed 28-06-2025].

[18] europarl.europa.eu. https://www.europarl.europa.eu/RegData/etudes/ATAG/2025/769509/EPRS_ATA(2025)769509_EN.pdf. [Accessed 30-05-2025].

[19] Fabris, A. *et al.* Fairness and bias in algorithmic hiring: A multidisciplinary survey. *ACM Trans. Intell. Syst. Technol.* **16** (2025). URL https://doi.org/10.1145/3696457.

[20] Ahuja, M. K. Women in the information technology profession: a literature review, synthesis and research agenda. *Eur. J. Inf. Syst.* **11**, 20–34 (2002).

[21] Help Wanted — upturn.org. https://www.upturn.org/work/help-wanted/. [Accessed 31-05-2025].

[22] Thorn, E. Resume Screening Tools: Types, Pros, Cons & Alternatives for 2025 — toggl.com. https://toggl.com/blog/resume-screening-tools. [Accessed 31-05-2025].

[23] frontiersin.org. https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2024.1352029/full. [Accessed 29-06-2025].

[24] All the Ways Hiring Algorithms Can Introduce Bias — hbr.org. https://hbr.org/2019/05/all-the-ways-hiring-algorithms-can-introduce-bias. [Accessed 30-06-2025].

[25] The Hidden Pitfalls of Traditional Resume Parsing Tech — recrew.ai. https://www.recrew.ai/blog/the-hidden-pitfalls-of-traditional-resume-parsing-the-disadvantages. [Accessed 02-07-2025].

[26] Overcome Top Resume Parsing Challenges: Proven Strategies — recrew.ai. https://www.recrew.ai/blog/top-7-resume-parsing-challenges-and-strategies-to-overcome. [Accessed 02-07-2025].

[27] Staab, R., Vero, M., Balunovic, M. & Vechev, M. Beyond memorization: Violating privacy via inference with large language models. In *The Twelfth International Conference on Learning Representations* (2024). URL https://openreview.net/forum?id=kmn0BhQk7p.

[28] Group vs. Individual Fairness in AI — lumenova.ai. https://www.lumenova.ai/blog/group-fairness-vs-individual-fairness/. [Accessed 02-07-2025].

[29] George John, P., Vijaykeerthy, D. & Saha, D. Verifying individual fairness in machine learning models. In Peters, J. & Sontag, D. (eds.) *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI)*, vol. 124 of *Proceedings of Machine Learning Research*, 749–758 (PMLR, 2020). URL https://proceedings.mlr.press/v124/george-john20a.html.

[30] AI and data privacy: What your company needs to know — usercentrics.com. https://usercentrics.com/guides/data-privacy/ai-and-data-privacy/. [Accessed 31-05-2025].

[31] van der Meijden, S. L. *et al.* Navigating fairness in AI-based prediction models: Theoretical constructs and practical applications (2025).

[32] Zhuang, F. *et al.* A comprehensive survey on transfer learning (2019). 1911.02685.

[33] Akhbardeh, F., Zampieri, M., Alm, C. O. & Desell, T. Transfer learning methods for domain adaptation in technical logbook datasets. In Calzolari, N. *et al.* (eds.) *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, 4235–4244 (European Language Resources Association, Marseille, France, 2022). URL https://aclanthology.org/2022.lrec-1.450/.

[34] Heakl, A., Mohamed, Y., Mohamed, N., Elsharkawy, A. & Zaky, A. ResumeAtlas: Revisiting resume classification with large-scale datasets and large language models (2024). 2406.18125.

[35] Yang, J., Soltan, A. A. S., Eyre, D. W., Yang, Y. & Clifton, D. A. An adversarial training framework for mitigating algorithmic biases in clinical machine learning. *NPJ Digit. Med.* **6**, 55 (2023).

[36] Vinod Bhatia, K., Capasso, M., Arora, P., Castillo, C. & Saldivar, J. Proxy discrimination risks in hiring: A qualitative analysis of a set of real CVs (2025).

[37] Findhr &x2013; Fairness and Intersectional Non-Discrimination in Human Recommendation — findhr.eu. `https://findhr.eu/`. [Accessed 28-06-2025].

[38] Resume Dataset — kaggle.com. `https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset`. [Accessed 03-06-2025].

[39] google-bert/bert-base-multilingual-cased · Hugging Face — huggingface.co. `https://huggingface.co/google-bert/bert-base-multilingual-cased`. [Accessed 02-07-2025].

[40] FacebookAI/roberta-base · Hugging Face — huggingface.co. `https://huggingface.co/FacebookAI/roberta-base`. [Accessed 02-07-2025].

[41] DistilBERT — huggingface.co. `https://huggingface.co/docs/transformers/model_doc/distilbert`. [Accessed 02-07-2025].

[42] Shifterator &x2014; Shifterator documentation — shifterator.readthedocs.io. `https://shifterator.readthedocs.io/en/latest/`. [Accessed 02-07-2025].

[43] ModernBERT — huggingface.co. `https://huggingface.co/docs/transformers/model_doc/modernbert`. [Accessed 02-07-2025].

[44] Papers with Code - Focal Loss Explained — paperswithcode.com. `https://paperswithcode.com/method/focal-loss`. [Accessed 02-07-2025].

[45] Ganin, Y. & Lempitsky, V. Unsupervised domain adaptation by backpropagation (2014). `1409.7495`.

Appendix

Table 6: ICT vs Non-ICT Job Category Classification for LiveCareer dataset

Job Category	**ICT Classification**
ACCOUNTANT	Non-ICT
ADVOCATE	Non-ICT
AGRICULTURE	Non-ICT
APPAREL	Non-ICT
ARTS	Non-ICT
AUTOMOBILE	Non-ICT
AVIATION	Non-ICT
BANKING	Non-ICT
BPO	ICT
BUSINESS-DEVELOPMENT	Non-ICT
CHEF	Non-ICT
CONSTRUCTION	Non-ICT
CONSULTANT	Non-ICT
DESIGNER	Non-ICT
DIGITAL-MEDIA	ICT
ENGINEERING	ICT
FINANCE	Non-ICT
FITNESS	Non-ICT
HEALTHCARE	Non-ICT
HR	Non-ICT
INFORMATION-TECHNOLOGY	ICT
PUBLIC-RELATIONS	Non-ICT
SALES	Non-ICT
TEACHER	Non-ICT

Table 7: ICT vs Non-ICT Job Category Classification for FINDHR dataset

Occupational Category	**ICT Classification**
English Categories	
Clerical support workers	Non-ICT
Service and sales workers	Non-ICT
Craft and related trade workers	Non-ICT
Associate professionals	ICT
Managers	Non-ICT
Professionals	ICT
Plant and machine operators and assemblers	Non-ICT
Elementary occupations	Non-ICT
Skilled agricultural, forestry, and fishery workers	Non-ICT
Armed forces	Non-ICT
English Categories (Detailed)	
Managers (managers, official, legislators)	Non-ICT
Professionals (health, legal, business, engineering, teaching)	ICT
Associate professionals (health, legal, business)	Non-ICT
German Categories	
Akademische Berufe (Naturwissenschaft, Ingenieurswesen, Gesundheit, Bildung, Verwaltung, Wirtschaft, Wissenschaft, Technologie, Medien, Jura, Sozialwissenschaft, Kultur)	ICT
Führungskräfte (diverse Bereiche)	ICT
Dienstleistungsberufe, Sicherheit und Verkauf	Non-ICT
Bürokräfte und verwandte Berufe (Administration, Kundendienst, Buchhaltung)	Non-ICT
Nichtakademische Techniker*innen und gleichrangige nichttechnische Berufe (Ingenieurswesen, Gesundheit, Verwaltung, Technologie, Wirtschaft, Soziales, Kultur)	ICT
Bedienung von Anlagen und Maschinen, Montage	Non-ICT
Spanish/Catalan Categories	
Personal de apoyo administrativo	Non-ICT
Treballadors de suport administratiu	Non-ICT

A Phase 1, anonymised resumes preprocessing

- Whitespace normalization: Conversion of multiple whitespace sequences to single spaces ensuring consistent tokenization across variable formatting styles.
- Contact information tokenization: Replacement of remaining email addresses and phone numbers with standardized tokens ([EMAIL], [PHONE]) preventing direct identification while maintaining document structure signals.
- URL standardization: Replacement of web addresses with [URL] tokens eliminating external identifier leakage.
- Structural preservation: Insertion of [NEWLINE] tokens maintaining document flow patterns relevant for linguistic analysis.
- Title detection: Recognition and standardization of professional titles as [TITLE_X] tokens preserving potential gender signals in controlled format.

B Phase 2, Data Preprocessing Architecture

PDF Processing and Text Extraction: Full PDF resume processing leverages *PyMuPDF* for layout-aware document parsing, preserving structural information critical for comprehensive resume analysis. The extraction methodology maintains document hierarchy through intelligent spacing recognition, preserving section boundaries and formatting cues facilitating subsequent semantic understanding. Text standardization procedures eliminate format-specific artifacts while preserving professionally relevant content through whitespace normalization, URL tokenization, and multilingual character encoding standardization supporting diverse linguistic content.

Multilingual Processing Pipeline: The FINDHR dataset's multilingual composition required translation processing to ensure model compatibility and performance consistency. Translation employs Facebook's *M2M-100* neural machine translation model through dedicated API processing, converting Spanish, German, and Catalan resumes to English while preserving professional terminology and semantic content. The translation process implements intelligent document chunking strategies optimizing translation quality while maintaining semantic coherence across resume sections. Language detection automation provides robust source language identification with fallback mechanisms for uncertain classifications. Translated content underwent manual validation procedures, ensuring translation quality and professional terminology preservation.

Protected Attribute Integration: For the FINDHR dataset, ID-based matching systems ensure precise integration of resume content with protected attribute information from survey data. This integration framework maintains data lineage throughout preprocessing.

C Transfer Learning Architecture

ModernBERT Foundation: The transfer learning architecture employs ModernBERT-base as foundational encoder.

ModernBERT's extended context length limits truncation challenges enabling comprehensive document processing, while Rotary Positional Embeddings enhance positional relationship understanding across long documents.

Pre-training Strategy: Initial training on the LiveCareer dataset leverages the larger scale (2,400 samples) to establish robust feature representations through comprehensive pre-training on the ModernBERT-base architecture. This foundational phase enables the model to learn generalizable textual patterns for resume classification before domain-specific adaptation to the FINDHR dataset's unique characteristics.

Hybrid Sampling Methodology: The pre-training methodology implements a two-stage hybrid sampling approach designed to address class imbalance while preserving semantic diversity. The hybrid strategy combines strategic undersampling of the majority class with targeted oversampling of the minority class, achieving a target ratio of 1:4 (minority:majority classes). Specifically, the undersampling component retains 60% of majority class samples through random selection without replacement, reducing dataset size while maintaining representative coverage. The oversampling component employs random sampling with replacement to duplicate minority class samples until achieving the predetermined target ratio of 0.25, ensuring sufficient statistical power for minority class learning without complete dataset balancing that might compromise majority class representation.

Focal Loss Implementation: Class imbalance mitigation employs Focal Loss optimization with a gamma parameter of 2.0, providing balanced focus on hard-to-classify examples without excessive emphasis that might destabilize training. The Focal Loss formulation implements dynamic re-weighting where well-classified examples contribute reduced loss values, while misclassified examples receive increased attention through the focusing parameter. This approach better addresses the challenge compared to standard cross-entropy loss which allows easily classified majority class samples to dominate the gradient updates, potentially overwhelming minority class learning signals.

Automated Class Weight Calculation: Complementary to Focal Loss, the training pipeline implements dynamic class weight calculation using inverse frequency weighting, where class weights equal total_samples / (num_classes × class_count). This automated weighting provides additional loss function adjustment that scales inversely with class frequency, ensuring minority class misclassifications contribute proportionally higher loss values during backpropagation.

Advanced Regularization Strategy: The pre-training phase incorporates regularization techniques including increased dropout rates (0.3 for both hidden layers and attention mechanisms), AdamW optimization with moderate weight decay (0.02) providing L2 regularization, and gradient clipping (maximum norm 1.0) preventing gradient explosion during imbalanced training scenarios. These techniques collectively prevent overfitting to the majority class while maintaining model generalization capabilities.

Memory-Optimized Training Architecture: Implementation employs gradient checkpointing for memory efficiency, enabling larger effective batch sizes within GPU constraints. The training pipeline utilizes stratified 5-fold cross-validation with early stopping based on F1-score monitoring, implementing patience mechanisms to prevent overtraining while identifying optimal model states for transfer learning.

Threshold Optimization Framework: The pre-training phase incorporates systematic threshold optimization that prioritizes high recall for minority class detection while maintaining minimum precision thresholds (0.7). This optimization searches across threshold values (0.1 to 0.9 in 0.05 increments) to identify decision boundaries that maximize minority class recovery rates, addressing the ICT underrepresentation in the dataset.

D Adversarial Debiasing Implementation Details

This section provides exhaustive implementation details for the adversarial debiasing pipeline described in Section 3.2.3. The technical specifications enable full reproducibility of the experimental design while complementing the theoretical framework established in the methodology chapter.

Network Architecture and Dimensions:

Base Architecture: ModernBERT-base (answerdotai/ModernBERT-base)

Backbone: 768-dimensional hidden states, 256 max sequence length.

Total Parameters: 110M (encoder) + 248K (classification heads)

Table 8: Component Architecture

Component	**Architecture/Operation**	**Input → Output**	**Parameters**
ModernBERT Encoder	Transformer (base)	[batch, 256] → [batch, 768]	∼110M
Job Classifier	Linear	768 → 2	
Gender Classifier	Linear → ReLU → Linear	768 → 256 → 2	197,378
Gradient Reversal Layer	Identity (λ scaling)	768 → 768	–
Gender Probe	Linear → ReLU → Linear	768 → 64 → 2	49,282

Adversarial Model Flow:

```
Input Text [batch, 256 tokens]
      ↓
ModernBERT Encoder [batch, 768]
      ↓
    Dropout(0.4)
      ↓
     -----------------
    ↓                 ↓
Job Classifier    Gradient Reversal Layer (lambda)
Linear(768->2)          ↓
    ↓             Gender Classifier
Job Logits        Linear(768->256->2)
                        ↓
                  Gender Logits
```

Training Hyperparameters:

Table 9: Experiment Configurations

Component	Learning Rate	Epochs	Batch Size	Optimizer	Weight Decay
CV Baseline (Frozen)	3e-4	2	32	AdamW	0.02
CV Baseline (Unfrozen)	8e-6	5	32	AdamW	0.02
Adversary Pre-training	1e-3	10	32	AdamW	0.01
Lambda Optimization	BERT: 5e-6 Gender: 5e-4	6	16	AdamW	0.02
Final Adversarial	BERT: 5e-6 Gender: 5e-4	15	32	AdamW	0.02
Final Baseline	1e-6	15	32	AdamW	0.1
Gender Probe	1e-4	15	32	Adam	-

Implementation of Pseudocode for Adversarial Training Loop:

```
# Phase 1: Baseline establishment
baseline_model = train_baseline(modernbert, findhr_data)

# Phase 2: Adversary pre-training
gender_classifier = GenderClassifier(768 -> 256 -> 2)
train_adversary(gender_classifier, frozen(baseline_model.encoder))

# Phase 3: Lambda optimization
best_lambda = grid_search(lambdas=[0.1, 0.5, 1.0, 2.0, 5.0])
```

```
# Phase 4: Adversarial training with progressive scheduling
for epoch in range(15):
    current_lambda = 2.0 * (epoch / 15)  # Progressive scheduling

    for batch in dataloader:
        # Forward pass
        features = encoder(batch.text)
        job_logits = job_classifier(features)

        # Adversarial path with GRL
        reversed_features = GradientReversalLayer(features, current_lambda)
        gender_logits = gender_classifier(reversed_features)

        # Combined loss
        job_loss = CrossEntropy(job_logits, batch.job_labels, class_weights)
        gender_loss = CrossEntropy(gender_logits, batch.gender_labels)
        total_loss = job_loss + gender_loss

        # Backpropagation (GRL handles gradient reversal)
        total_loss.backward()
        optimizer.step()

# Phase 5: Evaluation setup
extended_baseline = additional_training(baseline_model, 10_epochs)
probe_classifier = train_probe(frozen(extended_baseline.encoder))
```

Transfer Learning Adaptation: The adversarial debiasing architecture initializes from the best-performing model identified through cross-validation, leveraging pre-trained weights to maintain established feature representations. The implementation loads from the previous phase the highest F1-scoring fold model as the foundational checkpoint, ensuring robust baseline performance before debiasing modifications.

This way we generate the baseline model.

The domain adaptation employs a progressive unfreezing strategy to maintain training stability. Initially, the pre-trained encoder backbone remains frozen while only the classification head undergoes fine-tuning, allowing rapid convergence on the target domain without disrupting established feature hierarchies.

Subsequently, the entire architecture is unfrozen for end-to-end optimization, enabling deeper adaptation while preserving the beneficial inductive biases acquired during pre-training. This staged approach prevents forgetting of domain-relevant patterns while facilitating the integration of fairness constraints into the learned representations.

Pre-training the Gender Predictor The adversarial framework requires a competent gender predictor to provide meaningful gradient signals for the subsequent min-max optimization.

During this phase, the pre-trained encoder backbone and job classifier of the baseline model remains frozen while only the gender classification head undergoes training for three epochs. This ensures that the adversary develops some gender detection capabilities without disrupting the established ICT classification performance.

The classifier employs a three-layer fully connected architecture with ReLU activations and dropout regularization (0.4), transforming the 768-dimensional encoder features through hidden dimensions of 256 before binary classification. Pre-training utilizes an accelerated learning rate of $1e-3$ with AdamW optimization to establish adversary competence rapidly.

Gradient Reversal Layer Implementation The Gradient Reversal Layer is implemented as a custom PyTorch autograd function maintaining computational transparency during forward propagation (Identity operation) while reversing gradients during backpropagation. The lambda parameter controls reversal intensity, with higher values strengthening bias mitigation pressure.

Lambda Parameter Optimization Following adversary establishment, hyperparameter optimization determines the optimal trade-off intensity between job classification performance and gender debiasing effectiveness through grid search over lambda values $\lambda \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$. Each candidate lambda undergoes evaluation through adversarial training (6 epochs) using the pre-trained gender-predictor model as adversary.

The optimization employs differential learning rates with conservative encoder updates ($5e-6$) and accelerated adversary adaptation ($5e-4$) to maintain representational stability while enabling rapid convergence assessment.

Lambda selection utilizes a weighted scoring function that balances classification utility and bias mitigation:

$$\text{Score} = 0.6 \times F1\_\text{job} + 0.4 \times (1.0 - 2 \times |\text{Accuracy}\_\text{gender} - 0.5|)$$

where the best debiasing component rewards reduced gender predictability from learned representations while keeping reasonable ICT predictive performances.

Final Training Phase, Adversarial vs Baseline The final training phase implements adversarial optimization through dual model development and comparative evaluation on the complete available dataset.

Bias mitigation assessment employs a two-stage comparative framework.

First, the baseline model receives additional training for 15 epochs, reaching optimal performance in ICT job prediction without any debiasing component.

Then, baseline gender leakage is measured by training a probe gender classifier for 15 epochs on the baseline model's encoder representations to quantify residual gender information in the classifier's features.

Second, adversarial effectiveness is evaluated by testing the adversarial model's built-in gender classifier on the debiased encoder representations produced after 10 epochs of adversarial training. The model uses progressive lambda scheduling, gradually increasing from zero to the optimal lambda value to ensure stable convergence of the min-max objective.

Both models utilize the complete training dataset (combined train and validation sets) to maximize learning from available data.

Successful debiasing is indicated by a substantial reduction in gender predictability between these two measurements. Specifically, the adversarial model's gender classification accuracy should approach random performance (50%) while the baseline probe should display higher accuracy, indicating the presence of gender signals in the original representations.

Final evaluation quantifies the adversarial training effectiveness through dual-metric assessment: job classification performance measured via F1-score on ICT prediction tasks, and bias mitigation evaluated through gender predictability analysis.